The Acquisition–Authority Envelope for Open-World Scientific Discovery

Sze Chun Yiu
Independent Researcher
sze-chun.yiu@fysik.su.se

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Abstract

Open-world scientific discovery combines acquisition, screening and closure, yet these stages are usually evaluated separately. We define an acquisition–authority envelope that compares controllers under matched access, budget and authority contracts, and prove acquisition-ceiling, closure-factorization and donor-saturation results. Archived experiments preserve rather than average heterogeneous authority. A 390-task controlled index is descriptive; external screening does not establish superiority. In V7, six locked gates fail and the full candidate loses to its frozen u4 donor. V8 admits no residual, and V10 fails four gates. V13 validates a provider-native fibre-separating coordinate, but only 4/7 reviews support it. V15 binds a signed coherent snapshot containing 61 dataset blobs, without index parsing, census or performance. Its template error was never executed, and V15B corrects it only for a successor. Separately, density-normalized and query-conditional BEIR successors leave all five ArguAna stopping decisions unchanged, localizing that failure to a signal-inert rank-overlap marginal. Independent custody remains 0/3. Thus the framework identifies when new state or acquisition is necessary, but open-world benefit remains unconfirmed.

Keywords: open-world scientific discovery; information retrieval; deep research agents; acquisition envelope; scientific closure; systematic review; stopping rules.

1 Introduction

Scientific discovery agents increasingly combine retrieval, query reformulation, reranking, graph expansion, selective reading, state compression, contradiction handling and learned stopping. Treating any one component as "the discovery system" makes scientific comparison unstable: a new retriever can dominate the old acquisition layer without answering whether the assembled controller found the decision-relevant evidence, kept useful future routes alive, or closed the scientific task safely. We therefore ask a larger question: *what joint discovery frontier is attainable under a fixed source-access and resource contract, and what information is necessary for a point on that frontier to carry scientific-closure authority?*

The paper makes five contributions. First, it defines the *acquisition–authority envelope*, a non-compensatory frontier over decision-relevant discovery, scientific utility, future-search optionality, false closure, valid closure yield and total charged cost. This moves the unit of theory above individual retrievers and stopping rules. Second, it proves an acquisition-ceiling theorem that attributes an adverse outcome to route generation whenever the required evidence never entered the candidate pool. Third, it gives a necessary-and-sufficient state-factorization criterion for scientific closure and an indistinguishable-world lower bound for any controller whose observations conflate closable and materially open worlds. Fourth, it characterizes the maximally permissive robust completion and the necessary-and-sufficient completeness condition under which a registered obligation rule reaches it. Fifth, it instantiates the theory with earned route independence and question-conditioned read state, then places all archived positive, adverse and undetermined experiments at their exact authority levels rather than averaging them into one apparent win.

Let route r be characterized by a backend identity b_r, query-derivation process q_r, and capture/content identity policy c_r. A distinct route label is insufficient for independence. Evidence for route independence must be derived from the actual generation and capture process rather than asserted as a boolean.

2 Question-conditioned cumulative read state

A paper being previously encountered is weaker than the statement that it has been processed for the current research question, extraction schema and content version. Content identity is therefore held separately from question-conditioned read state. A changed question or materially changed content can legitimately reopen a source for processing without pretending that it is newly discovered evidence.

3 Route stopping versus task stopping

Route outcomes such as continue, reformulate, switch, budget-stop, unavailable and locally flat are separated from task-level scientific closure. A route may stop because its marginal yield is low while another route is still untried, unavailable or censored. Coverage diagnostics therefore stay advisory; they cannot promote global completeness.

This fail-closed rule is especially important for capture–recapture-style diagnostics. Adaptive search routes are not automatically independent capture occasions, and zero overlap cannot justify a confident bounded estimate of unseen literature.

4 Acquisition is not closure

The central design move in this paper is to separate two problems that are often collapsed into one policy. An *acquisition policy* decides what to search, retrieve, inspect, expand or reread next. A *closure authority* decides whether the evidence state is allowed to support route-local stopping or scientific task completion. Better acquisition can improve recall and reduce cost, but it does not automatically acquire authority to declare completeness. This separation makes strong external retrieval and stopping mechanisms composable rather than adversarial to the contribution of this paper.

Let x_t denote the observable search state at step t. We write an acquisition mechanism as

$$a_t \sim A(x_t),$$

where a_t may be a lexical query, a reasoning-aware query, a fielded Boolean filter, a graph-expansion action, a selective fetch, a reranking operation, or a request for another retrieval round. No novelty is claimed here for the internal form of A. Instead, acquired evidence updates both an evidence state E_t and a typed obligation state O_t. Closure is a distinct map

$$C(E_t, O_t) \in \{\textsc{continue}, \textsc{route-stop}, \textsc{open}, \textsc{cannot-check}, \textsc{task-close}\}.$$

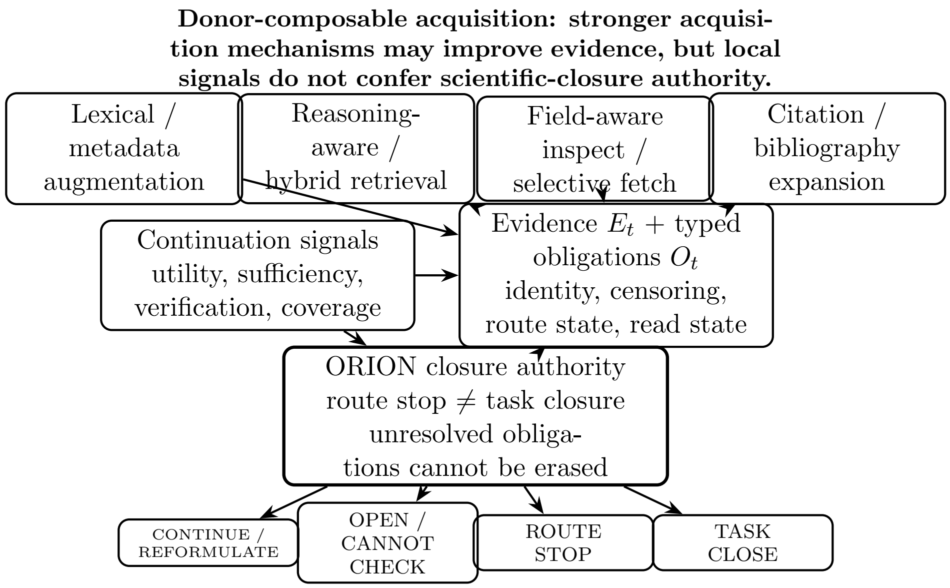

Figure 1: Acquisition mechanisms are intentionally composable: lexical/metadata augmentation, reasoning-aware or hybrid retrieval, field-aware selective fetching, citation expansion and continuation signals may all improve what is acquired. Their outputs update evidence and typed obligations, but a separate authority layer controls route stop, open and undetermined states and task closure. The figure therefore locates the residual at the meaning of acquisition outcomes rather than at a claim to have invented the strongest retriever.

A utility, relevance, confidence, sufficiency or coverage signal may influence A and may recommend a local stop, but it cannot by itself delete an unresolved obligation in O_t. This interface is where the residual contribution of this paper sits.

4.1 Absorbing strong acquisition mechanisms

The acquisition layer is intentionally donor-composable. SAGE shows that keyword-oriented agent queries can make strong lexical retrieval difficult to beat and that corpus-side metadata/keyword augmentation can further improve retrieval [9]. AgentIR shows that an agent's reasoning trace can be used jointly with its query as a retrieval signal [3]. SIEVE shows that fielded Boolean admission, structure-rich inspection and selective section fetching can improve the accuracy–context tradeoff [17]. Breadth-first bibliography expansion can recover literature missed by API-only retrieval, while also demonstrating why citation-list recall cannot be the only evaluation axis [14]. Lexical, metadata-augmented, reasoning-aware, field-aware and citation-expansion retrieval are therefore treated as candidate strengths to compose, not as inventions of this work.

The same rule applies to continuation and stopping signals. DeepControl uses marginal information utility to regulate continued retrieval [20]; decision-theoretic policies optimize screening stops against explicit costs and payoffs [7]; HALT uses claim–evidence coverage to verify whether expected evidence has been obtained [13]; MiCP applies conformal prediction to adaptive multi-turn reasoning to maintain a target coverage guarantee while permitting early stopping [22]; and structured Search-R1 stopping judgments explicitly encode sufficiency and remaining gaps [10]. These signals and guarantees can all be useful inputs to acquisition or route-local stopping. The question asked here is narrower: what are those signals permitted to certify when routes are dependent, unavailable, censored or only locally exhausted?

4.2 The authority residual

This decomposition yields four residual design claims that are separable from the mechanics above. First, route independence must be earned from backend, query-derivation and capture identity rather than from distinct tool names. Second, route-local stopping and task-global closure are different authority levels. Third, an unavailable or censored route remains a typed open obligation rather than becoming an empty result. Fourth, content identity and question-conditioned processing state must remain distinct so duplicate rediscovery does not inflate coverage while a genuinely changed scientific question can trigger justified rereading.

The resulting design is deliberately asymmetric. It is permissive about *how* evidence is acquired: a stronger retriever, query generator, expansion rule or verification signal can replace a weaker one. It is conservative about *what acquisition outcomes are allowed to mean*: local flatness, high estimated coverage, high confidence, apparent sufficiency or a provider failure cannot silently become proof of global completeness. This asymmetry is what permits the paper to absorb progress in retrieval without making its claimed authority boundary obsolete whenever a stronger retriever is published.

4.3 Empirical consequence

The decomposition also fixes the widening experiment. Candidate-generation methods should first be compared and composed under matched query/candidate budgets. Only after candidate recall is high enough to make stopping observable should the closure layer be challenged with direct ablations and strong stopping baselines. A positive external result must therefore show both acquisition quality and the incremental effect of typed closure authority; a cheaper stop that hides reachable relevant evidence is not a positive result. Conversely, a stronger donor retriever that improves acquisition while leaving the closure constraints intact strengthens the system rather than weakening the claim.

5 Formal mechanics

This section fixes the objects the evaluation measures. Every definition below is implemented in the system under test rather than assumed: work identity and read state, route independence and coverage, and route control each correspond to a module of the implementation, named in Section 14. Where a definition is deliberately conservative we state the failure it is conservative against, because the direction of the bias is what makes the refusal a safety property rather than a preference.

5.1 Work identity and content identity

Let R be the set of raw identifiers a route can emit (DOIs, arXiv ids, OpenAlex ids, local ids, URLs). Canonicalization is a map $\varphi : R \to \mathcal{S} \times V$ sending a raw identifier to a scheme-qualified key and a rendition version, so that $\varphi($`arXiv:2301.00001v1`$)$ and $\varphi($`arXiv:2301.00001v2`$)$ share a key and differ only in V: a version is a rendition of one work, not a second work. A declared scheme is treated as a claim rather than a credential; an identifier that does not match the shape of the scheme it announces resolves to UNKNOWN instead of entering the graph as a key that can never be fetched.

A *work* is the alias-closure of canonical keys. Given identity records with alias sets $A_1, \ldots, A_n$, works are the connected components of the graph on $\bigcup_i A_i$ that joins two keys sharing a record. Closure is computed as a projection over an append-only ledger rather than at write time, because an alias learned late can join two works recorded early; folding at write time would freeze the wrong partition.

Separately from work identity, each retrieved rendition carries a content digest c. Route overlap, deduplication and coverage are computed over content digests, never over retrieval identifiers: a source that re-mints identifiers would otherwise make two routes appear disjoint when they retrieved the same paper, which inflates every coverage quantity built on their overlap.

5.2 Question-conditioned read state

A read receipt is a tuple (w, c, σ, f): work, content digest, extraction schema version, and frame f — the scientific question the extraction was performed for. Given the receipt set $\mathcal{R}$, the read decision for a candidate (w, c, σ, f) is

$$\mathrm{read}(w,c,\sigma,f) = \begin{cases} \text{UNSEEN} & \nexists\, r \in \mathcal{R} : w_r = w \\ \text{CONTENT\_CHANGED} & \exists\, r : w_r = w,\ \nexists\, r : (w_r, c_r) = (w, c) \\ \text{NEW\_SCHEMA} & \exists\, r : (w_r, c_r) = (w, c),\ \nexists\, r : (w_r, c_r, \sigma_r) = (w, c, \sigma) \\ \text{NEW\_FRAME} & \exists\, r : (w_r, c_r, \sigma_r) = (w, c, \sigma),\ \nexists\, r : r = (w, c, \sigma, f) \\ \text{ALREADY\_READ} & \exists\, r \in \mathcal{R} : r = (w, c, \sigma, f). \end{cases}$$

ALREADY_READ therefore requires agreement on all four coordinates. This yields the distinction the evaluation depends on: a repeat extraction with all four coordinates equal is *duplicate processing* and is waste, while a repeat extraction under a changed content digest, schema or frame is a *legitimate reread* and is required work. Collapsing the two — treating "this paper has been seen" as "this paper has been read for this question" — is how a system reports coverage of material it only skimmed for a different purpose.

5.3 Earned route independence

A route capture is a tuple $\rho = (k, b, d, C)$: route kind, backend identity, query-derivation process, and the set C of captured content digests. For two captures ρ_i, ρ_j the independence predicate is

$$\mathrm{ind}(\rho_i, \rho_j) \iff k_i \neq k_j \ \wedge\ b_i \neq b_j \ \wedge\ d_i \neq d_j,$$

and the verdict otherwise names the shared cause (SAME_ROUTE_KIND, SHARED_BACKEND, SHARED_QUERY_DERIVATION). Independence is thus *earned* from the generation and capture process; a distinct route label carries no evidential weight.

Proposition 1 (Direction of the conservatism)**.** *Positive dependence between capture occasions biases unseen-mass estimators downward. Admitting a dependent pair therefore produces false confidence that search is near exhaustion, whereas refusing an independent pair only forfeits a diagnostic. The predicate is accordingly biased toward refusal.*

5.4 Coverage diagnostics are never authority

For an independent pair with captures of size n_i, n_j and overlap m_{ij}, the two-source estimate is $\hat{N} = n_i n_j / m_{ij}$, undefined at $m_{ij} = 0$. Zero overlap places no upper bound on the population, so the system returns a refusal with reason rather than a large number that would read as a measurement. When several independent pairs yield estimates, the reported value is $\max_{ij} \hat{N}_{ij}$: the largest population implies the most still unfound, which is the direction that fails safe. Every coverage estimate is marked diagnostic-only, and adaptive search is explicitly not a sequence of independent capture occasions — following up on what was just found converts singletons into doubletons and deflates the estimator toward "nearly done".

5.5 Route control state machine

Let a route hold attempts $a_1, \ldots, a_t$, each with retrieved digests D_i, novel digests $N_i \subseteq D_i$, cost, failure signatures and a backend-exhaustion flag. Marginal novelty of an attempt is $|N_t|/|D_t|$, and the flat streak z is the number of trailing attempts that succeeded, reported no exhaustion and returned no novel digest. Under a frozen policy with thresholds $\theta_{\text{ref}} \leq \theta_{\text{sw}}$, the decision is taken in strict priority order:

$$\delta = \begin{cases} \textsc{stop}\ \text{(budget)} & \text{remaining budget} \leq 0 \\ \textsc{stop}\ \text{(exhausted)} & a_t \text{ reports backend exhaustion} \\ \textsc{suspend} & a_t \text{ carries execution-failure signatures} \\ \textsc{switch} & z \geq \theta_{\text{sw}} \ \wedge \ \text{alternatives exist} \\ \textsc{reformulate} & z \geq \theta_{\text{ref}} \\ \textsc{continue} & \text{otherwise.} \end{cases}$$

Two separations in this table are load-bearing. First, backend exhaustion and transport failure are distinct states: the former is a statement about the index, the latter about the channel, and a system that merges them converts an outage into evidence of absence. The latter yields SUSPEND, a non-terminal state carrying an open obligation, not STOP. Second, each terminal decision carries typed reopen triggers (budget increase, backend recovery or refresh, material frame change, material finding on an alternative route); a route reopens exactly when an observed trigger intersects that set.

Proposition 2 (Route-stop non-certification)**.** *No individual route decision certifies task saturation. In the implementation every route decision,* STOP *included, has only route-local authority and carries an explicit declaration that it does not certify task saturation. Consequently, a route stop cannot be used by itself as a task-completeness certificate. A composition of route outcomes may support task closure only through a separately specified ensemble predicate whose route universe, material-obligation registry, provider/observability/custody guards, resource state and reopen triggers were fixed before the outcome.*

Proof. The route decision alphabet contains no task-closure authority, so projection of any one route output cannot emit a valid task certificate. The controller's task decision is instead produced by the separately typed closure map over the full ensemble state. If that map's registered completeness and validity guards hold, a composition can certify closure; if they do not, composition of route-local stops alone supplies no missing authority. □

Task closure is therefore a separate judgement over the route ensemble. It requires that every route in the prospectively registered route universe is in a terminal or explicitly reopenable state, that no suspended or unavailable route leaves an unmet obligation, and — by Proposition 2 — that no coverage diagnostic has been used as its authority. It also requires the registered provider, observability, custody and resource-validity guards to pass as one conjunction. An unavailable route remains an open coverage obligation and is reported as such; it is never converted into evidence that nothing further exists.

5.6 What the evaluation must therefore measure

The definitions above fix four measurable error families, each with a ground-truth condition that only a complete-gold world supplies: route-stop false positives (a route stopped while it could still reach unfound relevant items), route-stop false negatives (a route continued past its exhaustion point), premature task closure (task closed while a gold item was still reachable within remaining budget), and duplicate processing versus legitimate reread as separated by the four read coordinates. Route overlap, per-route marginal contribution and earned-independence status are computed over content digests. Metrics whose denominator is the gold set are reported only where that denominator is complete; the conditions are stated per metric in the frozen measurement plan.

6 The acquisition–authority envelope

The acquisition/closure split becomes substantially more useful when it is lifted from a design rule to a comparison theory. A scientific discovery controller has four coupled responsibilities: generate or select a route, acquire evidence through it, convert acquired material into decision-usable state while preserving useful future cues, and decide whether the task may close. A retriever improves only one part of this object; a stopping policy can save cost while making another part worse. This section defines the joint object and derives the conditions under which such local improvements can be composed into a scientifically stronger controller.

Mathematical conventions. Unless a probability law is explicitly introduced, exactness and factorization are pointwise, set-theoretic statements. Fibre constancy alone therefore proves a set map, not a measurable or computable implementation; a measurable factorization additionally requires a measurable quotient/factor map. Probabilistic statements are almost-sure under their declared laws and do not certify null fibres. Decision and action alphabets are nonempty. When total variation is used below, both laws live on one declared measurable transcript space and $\mathrm{TV}(P, Q) = \sup_B |P(B) - Q(B)|$, one half of the L^1 distance when densities exist.

6.1 Worlds, obligations and a non-compensatory frontier

Let $w \in \mathcal{W}$ be a scientific world and let h_t be the complete observable transcript after charged step t. The task contract assigns to w a set $\mathcal{O}(w)$ of *material scientific obligations*. An obligation may be a claim that needs evidence, a contradiction that needs both sides inspected, a dataset or code artifact needed to establish reproducibility, a route whose availability must be resolved, or a previously retrieved source that must be processed under the current question. Materiality means that resolving the obligation can change the predeclared decision, conclusion, uncertainty state or admissible closure state. It is not synonymous with relevance to a search query.

A controller π comprises route generation G, acquisition A, a question-conditioned state compiler M, and closure authority C. Under budget B, source-access contract $\mathcal{I}$ and task distribution P, let its outcome vector be (with the admissible policy class restricted to policies for which these coordinates are measurable and integrable)

$$Z_B(\pi, w) = (D, U, V, -F, Y, -K).$$

Here D is decision-relevant evidence discovery, U is final scientific utility, V is the value of search options preserved for prospectively hidden future questions, F is false task closure, Y is valid closure yield, and K is fully charged cost. Signs are chosen so larger is better except that F is a non-compensatory safety coordinate: a gain in report quality cannot purchase permission to hide a false closure.

For two outcome vectors $z, z' \in \mathbb{R}^6$, write $z \succeq z'$ for the coordinatewise product order,

$$D \geq D', \quad U \geq U', \quad V \geq V', \quad F \leq F', \quad Y \geq Y', \quad K \leq K',$$

and write $z \succ z'$ when $z \succeq z'$ and at least one inequality is strict. Thus a utility gain never compensates for worse false closure or cost; incomparability is retained as a scientific trade-off rather than collapsed by an unregistered scalar score. Let $q(\pi)$ collect the nonnumeric provider, observability, custody and contract-validity states of policy π. For a prospectively fixed guard registry $\eta = (\tau_F, \tau_Y, B, q_{\text{req}})$, define

$$\mathsf{G}_\eta(\pi) = 1 \quad \Longleftrightarrow \quad \mathbb{E}[F] \leq \tau_F,\ \mathbb{E}[Y] \geq \tau_Y,\ \mathbb{E}[K] \leq B,\ \text{and}\ q(\pi) = q_{\text{req}}.$$

An unbound or CANNOT-CHECK guard is not a pass.

Definition 1 (Acquisition–authority envelope)**.** For an admissible policy class Π,

$$\mathcal{E}_{B,\mathcal{I}}(\Pi) = \mathrm{cl}\left\{\mathbb{E}_{w\sim P}[Z_B(\pi, w)] : \pi \in \Pi,\ \mathbb{E}[K(\pi, w)] \leq B\right\}.$$

Its *guarded frontier* retains only points satisfying the prospectively fixed guard predicate G_η and then takes the $\succeq$-maximal points. Guard filtering and maximization are different operations: guarded attainable sets are monotone under donor addition, whereas their maximal-point sets need not be.

Definition 2 (Prospectively registered strict improvement)**.** Before outcomes, register a donor-complete class Π_0, the contracts $(P, B, \mathcal{I})$, guard registry η, primary coordinate j, strict margin $\delta_j > 0$, coordinatewise non-inferiority tolerances $\varepsilon \geq 0$, and a simultaneous one-sided uncertainty rule. Let $\mathcal{F}_0$ be the resulting guarded donor frontier. A candidate point z_c establishes *strict donor-complete improvement* only if its guards pass and, for every $z \in \mathcal{F}_0$,

$$z_{c,k} \geq z_k - \varepsilon_k \quad \text{for every benefit-signed coordinate } k, \qquad z_{c,j} \geq z_j + \delta_j,$$

with the sign convention above and with the registered simultaneous bounds, not point estimates, satisfying the inequalities. Safety coordinates may set no-harm tolerance to zero and always remain subject to G_η. Failure to bind the donor frontier, any guard, or the uncertainty rule is CANNOT-CHECK. A new nondominated trade-off may be reported as frontier extension, but neither that fact nor a win over a post-selected weak comparator licenses the word "superior."

This definition absorbs the strongest neighbouring components instead of opposing them. Lexical and corpus-side augmentation, reasoning-aware retrieval, fielded inspection and selective fetching, and citation expansion enlarge the action family of G and A [9, 3, 17, 14]. Claim–evidence coverage, marginal information utility, conformal continuation, decision-aware and evidence-aware stopping enlarge the policy and signal class [13, 20, 22, 7, 4, 6]. Finite source certificates, obligation graphs and question-conditioned memory enlarge M and the observable closure state [2, 21, 11, 1]. These are donor territory. The residual question is whether their composition reaches a better guarded frontier and whether its closure decision is identifiable from the state the product actually retains.

Proposition 3 (Donor-monotonicity)**.** *If $\Pi_1 \subseteq \Pi_2$ under identical task, access and budget contracts, then $\mathcal{E}_{B,\mathcal{I}}(\Pi_1) \subseteq \mathcal{E}_{B,\mathcal{I}}(\Pi_2)$. Consequently, adding a sound donor action, signal or memory operation cannot make the attainable envelope smaller, though it may leave the guarded frontier unchanged after its cost is charged.*

Proof. Every policy feasible in Π_1 remains feasible in Π_2; take the closure of their attainable outcome vectors. Intersecting both attainable sets with the same prospectively fixed guard predicate preserves inclusion. Taking maximal points afterward need not preserve inclusion, which is why the proposition is about attainable sets rather than frontier-point identity. □

The inclusion is an inclusion of closed attainable sets, not an inclusion of their Pareto-maximal or guarded-frontier points. A newly attainable point can dominate and thereby remove an old frontier point.

This elementary monotonicity result changes the novelty test. A new paper that improves retrieval should be absorbed into the comparator class. The theory is falsified, not protected, if the expanded donor class reaches every proposed candidate point under matched resources. Any strict-superiority statement must apply Definition 2; merely lying outside a plotted frontier, beating one convenient member, or choosing the improved coordinate after seeing outcomes is insufficient.

6.2 The acquisition ceiling: why seven emitted identifiers matter

Let G_w be a complete evaluator gold set for a fixed task world, $A_\pi(w)$ the union of gold-normalized identities ever emitted by any admissible route during a frozen run, and $S_\pi(w)$ the final submitted set. A downstream ranker, reader or synthesizer that is forbidden to invent evidence satisfies $S_\pi(w) \cap G_w \subseteq A_\pi(w) \cap G_w$.

Theorem 1 (Run-conditional acquisition ceiling)**.** *For any such downstream policy and any nonempty G_w,*

$$\mathrm{Recall}(S_\pi; G_w) \leq \frac{|A_\pi(w) \cap G_w|}{|G_w|}.$$

More generally, if obligations have nonnegative weights μ_o and each obligation can be discharged only by at least one member of evidence set E_o, then the unnormalized resolved-obligation weight is at most the total weight of obligations for which $A_\pi(w) \cap E_o \neq \varnothing$.

Proof. The submitted relevant set is a subset of the acquired relevant set, so its cardinality cannot be larger. For the weighted form, an obligation whose required evidence set is disjoint from the acquired pool cannot be discharged by a downstream operation over that pool; summing the remaining nonnegative weights gives the bound. □

The theorem separates candidate-generation failures from selection failures without assuming that a particular retriever is optimal. In the archived external adverse campaign, all routes emitted only 7 of the 229 gold identifiers; six were submitted and one was discarded. The run-conditional recall ceiling was therefore $7/229 = 0.030568$, and 222 gold identities were unavailable to every downstream selection rule on that capture. This does not estimate general web discoverability. It proves the narrower but decisive causal fact that reranking or synthesis could not repair that run. The adverse terminal and the separate provider-validity failure remain unchanged.

The same analysis exposes why matched request counts are not a matched scientific opportunity. Let $\xi(a, G_w)$ be the admissible exposure of action a: the prospectively validated opportunity, through the scorer-compatible interface, to emit identities in G_w. Equal $\sum_a 1$ does not imply equal $\sum_a \xi(a, G_w)$.

Proposition 4 (Call matching does not identify exposure matching)**.** *For every positive call budget m, there exist two m-call policies under the same nominal budget whose run-conditional acquisition ceilings are respectively one and zero. Hence call-count equality alone gives no nontrivial bound on the difference in discoverability.*

Proof. Take a one-item gold set. Give the first policy m calls whose admissible support contains the item with a deterministic first-call return, and give the second m calls to an interface whose output support is disjoint from the gold identity namespace. Both spend m calls; their acquired-gold fractions are one and zero. □

The archived campaign gave the lexical arm three calls to the only scoring-admissible backend and the governed arm one such call plus two inadmissible calls. Its three-versus-three request count was therefore not an exposure match. In a future protected study, exposure must be certified outcome-blind through interface-class witnesses or treated as CANNOT-CHECK; gold itself cannot be used as a worker-visible route oracle.

6.3 When can a product implement scientific closure?

Let $s : \mathcal{W} \to \mathcal{S}$ be everything a donor product makes available to its closure rule at a fixed decision time: acquired evidence, coverage and utility signals, route states, compiled memory and charged resource state. Write

$$\mathcal{Y}_C = \{\text{CLOSE}, \text{OPEN}, \text{CANNOT-CHECK}\}$$

and let $\kappa : \mathcal{W} \to \mathcal{Y}_C$ be the correct task decision under the declared scientific contract.

Theorem 2 (Closure factorization)**.** *There exists a decision map $g : \mathcal{S} \to \mathcal{Y}_C$ such that $\kappa = g \circ s$ if and only if κ is constant on every fibre $s^{-1}(z)$.*

Proof. If $\kappa = g \circ s$ and $s(w) = s(w')$, then $\kappa(w) = g(s(w)) = g(s(w')) = \kappa(w')$. Conversely, if κ is constant on every fibre, first define $g(z) = \kappa(w)$ for $z \in s(\mathcal{W})$ using any w with $s(w) = z$. Fibre constancy makes this definition independent of the chosen representative. Then extend g from $s(\mathcal{W})$ to all of $\mathcal{S}$ by assigning any fixed member of the nonempty alphabet $\mathcal{Y}_C$ to every $z \notin s(\mathcal{W})$. The extension does not affect $g \circ s$. □

The theorem gives an exact necessary-and-sufficient condition, not a claim that a particular architecture is intrinsically more expressive. Coverage, confidence and route-local stop signals are sufficient for task closure only when no two worlds with the same retained state require different task decisions. A state that drops the materiality or unresolved status of an unavailable route creates a mixed fibre and cannot be repaired by a different classifier over that same state. An information-equivalent product that retains the missing coordinate has pure fibres and can tie. This is exactly the boundary exhibited by the protected exact-contract battery: the available-route product fails on unresolved material routes, while the ideal typed product is decision-identical to the authority rule.

6.4 The price of an observationally unresolved world

The factorization obstruction has a quantitative version. Consider two task worlds w_0, w_1 whose correct decisions are respectively CLOSE and OPEN. Let P_0 and P_1 be probability measures on a common measurable transcript space $(\mathcal{X}, \Sigma_{\mathcal{X}})$, representing the complete observable transcript available to the controller before its decision. A randomized closure rule is a Markov kernel from $(\mathcal{X}, \Sigma_{\mathcal{X}})$ to the finite discrete alphabet $\mathcal{Y}_C$. Count CANNOT-CHECK as not yielding a valid closure in w_0 but not as false closure in w_1.

Theorem 3 (Indistinguishable-world lower bound)**.** *For every randomized closure rule, its false-closure probability F_1 in w_1 and valid-closure shortfall L_0 in w_0 satisfy*

$$F_1 + L_0 \geq 1 - \mathrm{TV}(P_0, P_1).$$

In particular, if the two worlds are observationally identical, then $F_1 + L_0 = 1$ and $\max(F_1, L_0) \geq 1/2$.

Proof. The decision rule is a binary test between P_0 and P_1 once OPEN and CANNOT-CHECK are combined as "do not close." The sum of its two testing errors is bounded below by one minus total variation. When $P_0 = P_1$, a transcript-dependent close probability has the same expectation in both worlds, so false closure plus failure to close the closable world equals one. □

This lower bound prevents a cosmetic conversion of every negative result into a positive terminal. If the decisive route is absent from the observable action support, no stopping heuristic can create the missing distinction. Scientific progress must instead do at least one of three things: invent an acquisition action that makes P_0 and P_1 distinguishable, retain a state coordinate that an earlier compiler erased, or introduce and validate a substantive world assumption that rules out one member of the pair. Otherwise a calibrated open or CANNOT-CHECK state is the only honest output.

6.5 Authority completion and its maximality

For a transcript h, let $\mathcal{W}(h)$ be the nonempty class of task worlds compatible with the declared observation and world contract, and let $A(w, h)$ say that closing at h is scientifically admissible in world w. The robustly safe histories are

$$\mathcal{H}_{\text{safe}} = \{h : A(w, h) \text{ holds for every } w \in \mathcal{W}(h)\}.$$

Theorem 4 (Maximally permissive robust completion)**.** *The rule $C^\star$ that closes exactly on $\mathcal{H}_{\text{safe}}$ has no false closure on any compatible world. Every other rule with that universal soundness property closes on a subset of $\mathcal{H}_{\text{safe}}$. Hence $C^\star$ is the unique maximally permissive universally sound close set, relative to the declared compatible-world class.*

Proof. Closure by $C^\star$ is admissible in every compatible world by the definition of $\mathcal{H}_{\text{safe}}$. If another universally sound rule closes at $h \notin \mathcal{H}_{\text{safe}}$, some $w \in \mathcal{W}(h)$ has $\neg A(w, h)$, contradicting soundness. Thus every sound close set is a subset of $\mathcal{H}_{\text{safe}}$. □ □

The theorem is exact but not directly computable unless the compatible-world class is represented adequately. The registered obligation rule is a conservative implementation candidate. Let $R(h)$ be the set of material obligations not yet independently discharged and let $V(h)$ say that every required task-contract, observability and provider-validity condition is established. Define

$$C^+(h) = \begin{cases} \text{CANNOT-CHECK}, & \neg V(h), \\ \text{CLOSE}, & V(h) \text{ and } R(h) = \varnothing, \\ \text{OPEN}, & \text{otherwise.} \end{cases}$$

It is sound whenever

$$V(h) \;\wedge\; R(h) = \varnothing \quad \Longrightarrow \quad h \in \mathcal{H}_{\text{safe}}.$$

It equals the maximal rule $C^\star$ if and only if the reverse implication also holds. Equivalently, the registry and validity predicate must be complete for robust closure, not merely sufficient. An adversarial-realizability condition—every unresolved material obligation has a compatible world in which closure is inadmissible—establishes that $R(h) \neq \varnothing$ implies $h \notin \mathcal{H}_{\text{safe}}$; it does not by itself prove that every safe history is recognized by V and R.

This distinction is the open-world boundary. A closed registry, a validated sampling model, independently curated hidden-route custody, or a declared CANNOT-CHECK state may justify the missing equivalence. Without it, C^+ is a sound lower approximation that may forgo valid closure, not a maximally permissive rule. Future acquisition, ranking, stopping or memory donors can be installed below C^+ without losing soundness when the sufficient condition is preserved; improvements that establish V, discharge obligations, or split a previously mixed closure fibre can move C^+ toward $C^\star$.

Corollary 1 (When acquisition enlarges closure authority)**.** *Adding a donor mechanism can expand discovery or reduce cost by Proposition 3. It enlarges the exact robust safe set only by changing the observation or compatible-world class so that a previously unsafe history enters $\mathcal{H}_{\text{safe}}$. For the operational rule C^+, this can occur by splitting a mixed closure fibre, discharging an obligation, establishing a missing validity condition V, or making a required action and terminal reachable under the registered budget/interface. The list is exhaustive only relative to those declared state coordinates.*

Corollary 2 (Comparator-adapter admissibility)**.** *Let T be a comparator's complete native transcript and let $a(T)$ be the normalized transcript supplied to the frozen evaluation. An adapter is lawful only if a is a prospectively fixed deterministic map. Thus under every declared world law $W \to T \to a(T)$ is a Markov chain, with no new query, source, world signal or unregistered action introduced by a. It must map native actions, costs, outputs and terminals into the frozen route/action/output namespace while preserving every registered terminal distinction. Other than prospectively fixed identity normalization, it cannot invent source identities or content; in particular, a native empty result, error or timeout remains non-evidence-bearing and cannot become acquired evidence, an obligation discharge, or task closure. Under these conditions every decision based on $a(T)$ also factors through T, so the adapter cannot split a native transcript fibre or lift the acquisition ceiling. It may only preserve or coarsen the comparator's information.*

Adapter lawfulness does not itself authorize closure. Soundness of a close terminal still requires the sufficient condition $V(h) \wedge R(h) = \varnothing \Rightarrow h \in \mathcal{H}_{\text{safe}}$, and equality with maximal robust authority still requires the reverse implication. Thus a native answer or end terminal may be recorded faithfully without being promoted to a P2 task-level CLOSE *terminal.*

This is the sense in which the proposed theory envelops its nearest work. Retrieval, coverage, information utility, stopping, obligation certificates and memory are all admissible mechanisms inside the envelope. Their scientific value is measured by the joint frontier they make reachable; their outputs do not acquire global authority merely by being numerically confident.

6.6 Safe residual envelopes and the donor-saturation boundary

Fix a source-bound finite family F, exact donor d, residual r and a predeclared conjunction $\mathcal{K}(F)$ of CRE20, R@10, WSS95, sign, harm and absolute-work-saving gates. Let B_F collect source, population, implementation and custody bindings, with every unbound predicate treated as false. The certificate-switched policy is

$$\pi_r^{\text{env}} = \begin{cases} e_r, & B_F = 1 \text{ and } x(r) \in \mathcal{K}(F), \\ d, & \text{otherwise.} \end{cases}$$

Thus an unavailable or rejected residual is not imputed as a tie or zero; it executes the exact donor.

Theorem 5 (Simultaneously certified residual embedding)**.** *Suppose the family, donor, residual, endpoints and gates are fixed before candidate outcomes. If a simultaneous $(1-\alpha)$ confidence set for the complete gate vector is contained in $\mathcal{K}(F)$, then the probability of admitting a target vector outside $\mathcal{K}(F)$ is at most α. Moreover, the attainable set of the certificate-switched class contains the donor attainable set, and every rejected or unbound residual executes d.*

Proof. False admission requires the true vector to lie outside $\mathcal{K}(F)$ while the confidence set lies inside it, which is possible only on simultaneous noncoverage. Donor containment follows by construction, and fallback is the second branch of the switch. □

The conjunction is noncompensatory: a sufficiently large gain on one endpoint cannot buy a failed coprimary, sign or harm gate. In particular, V10's positive mean WSS95 cannot repair its failed CRE20/R@10 magnitude and sign gates.

Now let $s_D(w)$ be all information and retained state available to a donor class Π_D in world w, and let $\mathcal{A}_D(w)$ be its admissible action and acquisition support under the matched contract. Call Π_D saturated if it contains every allowed policy measurable from s_D that uses only $\mathcal{A}_D$.

Theorem 6 (Donor-fibre saturation boundary)**.** *Under the same task, access, budget and authority contract, a candidate whose action is constant on every fibre of s_D and lies in $\mathcal{A}_D$ belongs to a saturated Π_D. Therefore strict outside-envelope separation requires at least one of: a signal or retained state that separates a donor fibre, an acquisition action outside donor support, or proof that the declared class was not saturated. Changing the cost, access, world or authority contract changes the estimand and does not establish dominance under the original contract.*

Proof. Fibre constancy gives a well-defined factorization $c = g \circ s_D$. Matched action support and contract guards make this an admissible donor policy, which saturation includes. The separation statement is the contrapositive; a contract mismatch instead violates the comparison premise. □

Finite residual failures do not establish saturation or universal donor optimality: absent a cross-world restriction, two extensions can agree on all observed reviews while a new-family residual fails every gate in one and passes them in the other. The title-emphasis transform is deterministic from fields already in donor state, so it adds no information; V10 rejects that one algorithmic transform, not every possible same-information learner.

6.7 A failure-derived candidate: obligation-directed adaptive exploration

The theory also specifies a materially stronger acquisition candidate rather than treating fail-closed authority as a substitute for finding evidence. Let $m_t(o)$ be the posterior unresolved mass of material obligation o, let $Q_t(a)$ be the prospectively evaluated option value that action a preserves for hidden future questions, and let $J_t(a)$ be the expected reduction in closure-fibre ambiguity—the probability mass of closable/open worlds that the action is expected to separate. An *obligation-directed adaptive exploration* (ODAE) policy chooses from route invention, retrieval, processing, contradiction resolution, state preservation/recovery and closure verification by a charged index of the form

$$a_t \in \arg\max_{a \in \mathcal{A}(h_t)} \frac{\mathrm{LCB}_t\left[\Delta \sum_{o \in R(h_t)} \mu_o m_t(o) \mid a\right] + \lambda Q_t(a) + \beta J_t(a)}{\mathrm{cost}_t(a)}.$$

The lower-confidence term prevents an uncertain route estimate from being treated as established gain; Q_t prevents current-question compression from destroying future route cues; and J_t values actions that make the closure decision identifiable even when their immediate document count is low. If no action has validated support through the frozen interface, the controller emits CANNOT-CHECK rather than assigning that route zero value.

Generic value-of-information, active search, coverage estimation and adaptive stopping are donor-owned. The proposed residual is their non-compensatory coupling to material scientific obligations, option preservation and closure fibre separation under one charged budget. ODAE is a prospective policy family, not an executed result or current general capability. Its direct falsifier is a donor-complete controller that reaches the same or a better guarded frontier on fresh tasks. Its causal tests must separately vary (i) route support, (ii) retrieved-but-unprocessed evidence, (iii) contradiction sides, (iv) future-query optionality and (v) closable clean cases; improvement on only the family that inspired the policy is insufficient.

6.8 A prospective cross-arena discriminator

The theory converts the historical adverse terminals into explicit research problems rather than deleting them. The 7/229 exposure result demands route invention and admissible-support estimation before reranking. The one-third scoring-eligible exposure mismatch demands source-interface matching rather than call-count matching. The retrieved-but-unprocessed and future-query failures demand option-valued state preservation. Provider-invalid and unresolved-route failures demand independent observability custody and an authority-complete state. Clean closable tasks remain a required counterweight against an always-open policy.

A separately frozen successor gives these problems a common empirical target. It specifies 768 source-disjoint task-world clusters—192 in each of four strata—across four arenas, identical admissible route exposure, three frozen primary comparators, an intersection requirement over every contrast, and a joint power target of 0.90 under its declared planning model. That design is a protocol, not a result. Archived baseline-task-world reproduction, four source-disjoint arena snapshots, a capability-matched external comparator, independent evaluator/gold custody, and live response/cost receipts are still required. Until those bindings exist and the protected campaign runs, the general cross-arena envelope-superiority hypothesis remains untested and the historical manuscript terminal remains unchanged.

7 Methods

The design choices in this section were fixed before the result-bearing offline campaign. Post-outcome edits to this prose are limited to correcting execution status (for example, saying that an external benchmark has not yet been run) and do not change a hypothesis, task family, system, metric, budget, exclusion rule, margin or decision rule. Machine-readable artifacts govern rather than this narrative: a design freeze, a measurement plan, a statistical plan, an external-artifact access audit, the controlled-world freeze and a prospective execution binding. Each is archived, and Section 14 names where.

7.1 Three-valued outcomes

Every decision this paper measures is three-valued rather than binary. A task may be determined complete, determined incomplete, or left *undetermined*; a route may be determined exhausted, determined to have more to give, or left undetermined. "We could not determine this" is recorded, reported and counted separately from "we determined this is false", and the two are never merged. The distinction is the point: a provider outage, a missing credential or a censored route produces an undetermined outcome, and an undetermined outcome is not evidence of absence. Throughout the paper an outcome described as *undetermined* is this third value, carried in the evidence record and in every downstream count, never silently resolved to a negative.

7.2 Prospective discipline

The protocol is frozen and versioned before execution. A result-bearing execution is admissible only after subject code, data, evaluator, system identities, budgets and repeat seeds are bound. For the completed offline companion, the run manifest records that no outcome was accessed before the freeze, together with the exact subject revision, content hashes, 14 system identifiers, three repeat seeds and host-owned resource limits. The bound subject revision passed the full test suite before the first aggregate outcome snapshot was recorded. Publication summaries and manuscript prose may be generated after outcomes, but they may only project the frozen records; they cannot change the scientific decision rules.

7.3 Evaluation environments

Three environments are distinguished because no single one supplies both a complete denominator and open-world realism. The first supplies the only full system comparison with a complete denominator; later bounded external probes have narrower acquisition or screening authority and are reported separately.

Offline complete-gold controlled index. Recall is only measurable where the relevant set can be enumerated. The controlled index is therefore fully synthetic and self-generated. Each topic has a materialized gold set under the frozen generator, and tests recompute the world and tasks from the recorded seed and compare all committed hashes rather than trusting a hand-authored denominator. Discoverability is heterogeneous: relevant items are partitioned across lexical, semantic, citation, reformulation and restricted/provider routes; two route labels deliberately share a backend; duplicate locators share content identity; revisions share identity but change digest; extraction-question-shift cases require legitimate rereads; and unavailable-route cases distinguish transport censoring from absence. Budgets are below exhaustive probing, so route allocation and stopping remain observable variables.

The completed suite has 390 tasks over five case families, drawn from 78 topics over 1,210 documents, frozen at suite fingerprint `2f6936ba52fb`. Every candidate uses the same host-owned session. The session alone can access the world, meters queries and reads, records route stops and transport states, and evaluates claims after the candidate returns. The candidate receives only a public task view and label-free retrieved records; protected gold and the relevance rule are never passed through the system interface.

External benchmark tasks. AutoResearchBench provides Wide and Deep scientific-discovery tasks [19]. The access audit establishes a narrower fact than an external result: Wide's official exact IoU/recall/precision path scores normalized arXiv identifiers without a credential, while its sampled maximum-IoU-at-k family is noisy because upstream sampling is unseeded. Deep's official title-matching evaluator needs an OpenAI-compatible judge. The Wide scorer was exercised on synthetic scorer input to verify that credential independence; no matched Wide result against a baseline is archived in this revision, and the scorer is not claimed to be vendored here.

SAGE is retained as neighbouring work and as a lexical-baseline warning [9], but is not executed as if its published setting were available: the paper-linked repository is data-only at the audited revision and provides neither an official runner nor a licence. Any substitute corpus, runner or weighting rule would be a declared deviation rather than the SAGE system. MetaSyn's separation of retrieval from screening is adopted as a measurement constraint [18]; its credential-free BM25 plus public-protocol screening probe subsequently completed on all 86 released test reviews and is reported as a bounded retrieval/screening result, not as the full P2 system. AgentSLR remains a protocol-driven systematic-review neighbor/baseline family rather than an official evaluator dependency [12].

Frozen external retrieval snapshots and live-provider campaign. The prospective external design freezes a gold-blind candidate pool per query before comparing systems, then runs a smaller predeclared live subsample. Pool coverage must be reported as a shared ceiling because a gold-blind pool can omit gold. Live requests are designed to retain URL/parameters, query derivation, timestamp, raw response and typed transport outcome. The capture machinery and manifest exist and have been exercised, but the final result-bearing external pool and live campaign have not been executed in this revision. Therefore no external cost, latency or recall result is inferred from the capture audit.

7.4 Systems under test

The implementation evaluated here is the ORION discovery layer; it is the artifact under test, and the rest of the paper calls it the *governed system* so that the claims read as claims about the mechanism rather than about a product. The frozen offline companion compares the governed system with six confirmatory comparators: a BM25 keyword retriever, a dense retriever, a sparse–dense hybrid, a one-pass retrieval-augmented pipeline, a single-route agentic searcher, and a protocol-driven systematic-review comparator. A seventh comparator, an adaptive multi-route exploratory searcher, was kept explicitly exploratory and was not retrofitted into the frozen confirmatory baseline set. The registry that binds each of these to its frozen system identifier is archived with the campaign.

These controlled-index systems are mechanism comparators, not claims to replicate a production BM25 engine, a particular embedding model or an external agent implementation. They consume the public synthetic route interface under identical budgets so that the variable under test is route/read/stopping governance. Their relevance screen sees only the public question and retrieved bibliographic text. The authored synthetic text makes screening deliberately easy; the experiment is not presented as evidence about classifier quality.

Six frozen ablations remove exactly one governance mechanism or authorize one forbidden transition: route-independence checking, the question coordinate in the read ledger, route/task stop separation, unavailable-route open state, coverage-diagnostic non-authority, and content-identity deduplication. All use the same session and budgets as the governed system.

7.5 Identity, reads and stopping

A read is keyed by content identity, content digest and extraction question in the full system. A new question can therefore reopen unchanged content for a legitimate reread, while mirrors with the same content identity/digest do not become new evidence. The corresponding ablations remove the question coordinate or replace content identity with document locator to expose suppression or duplicate work.

Route stop and task closure are separate actions. A route may stop after its public probe space is exhausted or after a typed transport outcome, but task closure is invalid if relevant material remains reachable within budget. An unavailable route stays censored and open rather than being converted to an empty result. Coverage diagnostics may inform a route policy but have no authority to certify global completeness.

7.6 Measurements

The offline evaluator reports denominator-valid complete-gold recall, precision, missed-gold count, route use, route/content overlap diagnostics, marginal relevant gain after the first route, premature task closure, censoring, duplicate processing and legitimate rereads, plus host-owned resource counters. Claims are normalized by content identity rather than locator so mirrors cannot inflate recall. Rich per-run artifacts retain route calls, reads, stop events, authority flags and raw-artifact hashes; the publication summary binds the full 16,380-record and rich-artifact sets by SHA-256.

The external measurement plan additionally defines official Wide/Deep scores, screening metrics and ranking-only metrics where their denominators and task semantics support them. Those definitions do not imply that an external result exists.

7.7 Statistics and authority

The statistical unit is the task. Repeated stochastic runs nest within task and system and never count as additional tasks. The frozen plan defines paired bootstrap comparisons, Wilson intervals for standalone binary rates, superiority/non-inferiority margins, a cost ceiling and Holm correction for the external confirmatory analysis. It also defines an authority ladder. The completed offline suite has $n = 390$, which reaches the precision tier the plan committed to in advance—registered in that plan as `TIER_B_committed` and requiring $n = 385$—at an achieved half-width of 0.0496; because that half-width exceeds the frozen superiority margin, the primary offline result carries the plan's mandatory underpowered label. The three deterministic repeat seeds verify stable execution; they do not manufacture additional tasks. We therefore report descriptive absolute differences and mechanism diagnostics, with no promoted superiority decision from the offline companion.

For a future external confirmatory result, the frozen primary rule requires the family superiority margin of 0.03 under matched resources with the premature closure safety guard no worse than 0.02. Non-inferiority uses the predeclared 0.01 margin and is inadmissible above the 1.25 consumed-cost ratio. Secondary families use Holm correction. These rules remain prospective because no external system result is reported here.

7.8 Failures, exclusions and contamination

Candidate failures, malformed outputs, budget exhaustion, abstention and typed transport failures remain outcomes. Provider unavailability cannot become evidence of absence. The frozen evaluator uses an explicit failure taxonomy and terminal precedence; the publication table therefore warns that a lower-priority mechanism can be masked by a higher-priority terminal classification such as premature closure. Public benchmark questions and gold identifiers also create a contamination risk for future live search. A credential-free search-time self-exposure diagnostic over the frozen 400-task AutoResearchBench Wide candidate run is archived with the external results and reports a task rate of 0.0. That diagnostic does not claim absence of indirect, training-time, or live-provider contamination.

7.9 Decision rule

The paper-level rule is fail-closed. The offline companion can support a mechanism claim only at descriptive authority. External discovery superiority requires an archived denominator-valid external result under the frozen matched resources and safety rule. If that evidence is absent, blocked or undetermined, the claim is narrowed rather than promoted by analogy from the synthetic world.

8 Results

8.1 Authority of the result

Three result-bearing evidence families are complete in this revision, with different authority. First, the frozen offline complete-gold companion contains 390 tasks, 14 systems (the governed system, six confirmatory baselines, one explicitly exploratory comparator, and six ablations), and three predeclared repeat seeds, for 16,380 normalized run records. The systems are deterministic; the repeats therefore test harness stability and do not increase the statistical unit beyond 390 tasks. The precision tier the frozen statistical plan committed to in advance requires 385 tasks, which the assembled 390 tasks reach at an achieved half-width of 0.0496 for a standalone rate. Because that half-width still exceeds the frozen superiority margin of 0.03, the plan's underpowered label attaches to the primary offline result: no superiority decision from the offline companion is promoted to a paper-level claim.

Second, the credential-free MetaSyn probe is a bounded external retrieval/screening result. It completed all 86 released test reviews using the pinned MetaSyn BM25 title+abstract retriever followed by the predeclared public-protocol screening rule, and was scored by the pinned official ID-only evaluator. That probe is not the complete multi-provider system and is therefore reported separately from the paper's still-open matched external comparison against a baseline.

Third, the archived public-screening transport campaign reports fixed-pool screening transport evidence on sources outside the SYNERGY development family. The one-pool Zenodo V1 binding failure, V2 post-schema-repair result and post-V2 active-comparator audit remain separate identities. A later SWIFT successor retained two additional population-mechanics failures and then bound five review decisions with 96,241 canonical rows. Its V3 unweighted review-level candidate-minus-u4 means were −0.021717971642 for recall at 10% screened and −0.050396333672 for WSS@95; u4 was better on both endpoints in all five reviews. The primary and relative-work-saving gates failed, while harm and absolute work saving passed without compensation. These screening studies inform downstream selection within acquired pools; they are not evidence about open-world route acquisition or closure, and V3 remains public-development evidence rather than population transport or confirmation. The subsequent KIFMS V6 packet is not a fourth performance family: it binds a lawful 14-review source, a 4,934-row content-disjoint population and a continuous recall–effort coprimary before exact V6 outcome access, but executes no arm. Its independent custody, label-semantic/class, harm, work-saving and performance coordinates therefore remain CANNOT-CHECK; V5 remains the latest independently unopened comparative screening result. V7 then uses a new hash-frozen transparent-public identity to execute those exact public data without claiming external custody. All 14 source files, 4,934 canonical rows and both-class checks pass. The learner/balancer means are +0.084472 CRE20, +0.148820 R@10 and +0.119813 WSS95, while the locked sign and harm gates fail; the full candidate loses to u4 at both R@10 and WSS95. V7 is therefore an adverse public-development mechanism result, not confirmation or promotion. V8 then tests whether a residual can safely envelop u4 using nested leave-one-review-out development. Exact u4 reproduces in 14/14 reviews, but 12/14 outer folds select fallback and both residual activations harm their held out review. Aggregate deltas versus u4 are −0.002009 CRE20, −0.028274 R@10 and −0.000991 WSS95, with worst R@10 −0.333333; no residual is admitted. V9 prospectively freezes a seven-review, source-disjoint title-emphasis test but stops before vectorization when one normalized title–abstract identity has two labels. V10 resolves only the identity semantics: provider records 1003 and 1018 have distinct author/DOI hashes and no provider duplicate link, so both are retained without choosing a label. Exactly one three-arm execution on 21,897 rows takes 357.789818 seconds. Candidate-minus-u4 means are +0.000779993 CRE20, −0.005382681 R@10 and +0.001765467 WSS95; positive CRE20 and R@10 occur in 4/7 and 1/7 reviews, while worst R@10 is −0.044444444. The two coprimary magnitude gates and both sign gates fail, so the residual is discarded and exact u4 remains the fallback. Positive WSS95, harm and absolute-work-saving gates do not compensate for those failures.

V11 introduces no new empirical run. It proves a conditional safe-residual envelope: exact donor fallback is embedded, simultaneous noncompensatory certificates bound false admission at their joint noncoverage level, and finite V8/V10 failures do not identify universal u4 optimality. Strict same-contract separation from a saturated donor-complete class requires a fibre-separating signal or retained state, acquisition support outside the donor, or proof that the declared donor class was incomplete. A changed cost, access, world or authority contract changes the comparison rather than proving dominance under the original one.

V12–V15 then tested the first of those ascent routes without opening a performance outcome. V12 found different provider-native keywords on two records tied under exact u4 model text, but its normalization omitted V10's trailing separator for an empty abstract. The exact V10 content-identity gate therefore failed, leaving the keyword difference as post-gate diagnosis only. V13 imported the exact V10 identity and passed all eight frozen bindings. It thus establishes one non-simulated provider-native coordinate that separates two records within an exact-u4-text fibre. Only four of the seven V10 reviews are keyword-capable, however, so no keyword-only residual can meet the unchanged six-of-seven sign gates and no performance run was authorized.

V14 sought a source-disjoint keyword-capable family but stopped at the source identity gate. The frozen commit resolves to a tree and index blob that are not the frozen expected blob; those expected bytes belong to a later commit, which was not substituted. All five identities are listed in the availability statement. V15 repaired the source-state binding without crossing into population or outcome access. The provider-valid signed commit, tag and recursive tree coherently expose the exact 22,135-byte index entry and 61 same-snapshot dataset CSV blobs. This is source-state availability, not an eligible census or screening benefit. The exact snapshot's root licence is MIT, while current repository metadata reports CC0; these authorities are kept separate. The exact-index attestation endpoint returned 404, independent custody remained open, and zero index, census, ranking or metric run occurred.

The offline campaign is summarized by a single immutable publication record. It binds the full 16,380-record set by SHA-256 `c6430a651810f8e7a794aa0c1091794963c43389a1e5080c02c2807a2fc2c574` and the corresponding rich-artifact hash list by `d851f168faaf50969198180ecc61a6ac361c72556ceeb992f011af307dd00c37`. The exact run manifest was committed before the first outcome snapshot. The mechanism projection used for Figures 4–5 is archived separately; it records the same two source digests and is regenerated from host-owned traces rather than copied from manuscript numbers. The MetaSyn result and its exact integer error decomposition are bound by two further archived evaluator records. Section 14 names every one of these artifacts and where it lives.

8.2 Controlled-index comparison

The governed system attains mean complete-gold recall 0.979487 and precision 1.0. It passes 260 of 390 tasks, fails on 59 tasks after honestly exhausting its route-call budgets without certifying completeness, and leaves 71 tasks undetermined because a route holding unretrieved relevant material

System	Recall	Premature closure	Pass	Cannot check	Fail
Governed system	0.979487	0.00	0.67	0.18	0.15
Protocol-driven SLR	0.666667	1.00	0.00	0.00	1.00
Sparse+dense hybrid	0.555556	1.00	0.00	0.00	1.00
One-pass RAG	0.555556	1.00	0.00	0.00	1.00
BM25/keyword	0.333333	1.00	0.00	0.00	1.00
Dense-route comparator	0.333333	1.00	0.00	0.00	1.00
Agentic single route	0.333333	1.00	0.00	0.00	1.00

Table 1: Controlled-index descriptive comparison. Rates are over 390 tasks; the failure column on the governed system's row is honest budget exhaustion, not premature closure. The frozen plan's underpowered label applies to the primary offline comparison, so no superiority interval or decision is reported here. A prospective external Wide campaign was executed but failed its frozen transport validity gate; its outcome is retained separately as undetermined.

becomes unavailable; it never certifies a censored task complete. Its premature-task-closure rate is 0. The strongest of the six protocol-frozen confirmatory baselines is the protocol-driven systematic-review comparator, with mean recall 0.666667. That comparator claims task completion on all 390 tasks while reachable gold remains, so its premature-closure rate is 1.0. The descriptive recall difference is therefore +0.31282, with a descriptive premature-closure difference of -1.0. These numbers are evidence about the authored controlled world, not an estimate of real-web superiority.

The two systems are identical through their first three search calls in this authored world. The protocol-driven comparator then reaches mean recall 0.666667 by query 7 and remains at that value through query 12. The governed system is at 0.757835 by query 7 and continues to gain, reaching 0.979487 by query 12. Figure 3 therefore localizes the descriptive gap to evidence gained after the early common search prefix rather than to an initial retrieval advantage.

8.3 Mechanism ablations

The safety ablations fail in the directions the frozen design intended them to probe. Removing the route-independence check reduces mean recall to 0.444444 and produces premature closure on every task. Allowing a route stop to certify task closure yields recall 0.333333 with the same 100% premature-closure rate. Letting a coverage diagnostic acquire stopping authority yields recall 0.555556 and again closes every task prematurely. These are negative controls: their purpose is to demonstrate that the forbidden transitions are observable, not to serve as competitive systems.

Removing content-identity deduplication is less catastrophic but measurably wasteful in the controlled world: mean duplicate-processing rate rises from 0 to 0.31453, 193 of 390 tasks exhaust the read budget, pass rate falls from 0.666667 to 0.369231, and mean recall falls to 0.939031. Removing the question coordinate from the read ledger leaves aggregate recall unchanged at 0.979487 but changes the extraction-question-shift mechanism exactly where expected: the mean count of legitimate rereads on that family falls from 3.666667 to 1.0. This is a mechanism result, not evidence of a recall gain from the ledger. Finally, removing the unavailable-route open state leaves mean recall numerically unchanged but turns the 71 censored undetermined cases into asserted completions and premature-closure failures. The distinction is therefore one of epistemic safety rather than headline recall.

The lexical route contributes 3.0 first relevant identities per task on average. Semantic and citation routes each contribute 2.0 additional identities, reformulation contributes 0.848718, and the restricted route contributes 0.966667; the last mean is below one because tasks that end censored or budget-exhausted before the restricted route is consulted contribute zero. Contribution and independence are deliberately separate variables: reformulation adds relevant content but shares the lexical backend and therefore cannot be promoted to an independent capture occasion.

Figure 5 exposes why numerical disjointness is not equated with route independence. In particular, lexical and reformulation have zero mean content overlap here yet are still rejected as independent because they share a backend. Conversely, lexical and semantic have nonzero mean overlap (0.07557) but remain structurally independent because neither backend nor query derivation is shared. The independence decision is therefore made from capture construction, while overlap remains a measured diagnostic.

8.4 MetaSyn external retrieval and screening probe

The corrected MetaSyn probe completed all 86 released test reviews with no failed review. Under the pinned official ID-only evaluator, macro retrieval recall is 0.7485 and retrieval precision 0.0627. Conditional retention among retrieved reference items is 0.7224; final inclusion recall is 0.5651, inclusion precision 0.1426 and inclusion F1 0.2033. The evaluator reports post-retrieval loss 0.1834 and screening accuracy 0.7228. Mean candidate output is fixed at 200 retrieved and 60 included items per review by the declared probe budget.

The evaluator-side integer ledger makes the stage loss concrete. Across the 86 reviews there are 1,677 reference-article links. The top-200 retrieval pools contain 1,079 of those links, so 598 are retrieval false negatives. Of the 1,079 retrieved reference links, 736 survive the deterministic screen and 343 are lost post retrieval. The final included sets therefore miss 941 reference links in total. Gold is loaded only on the evaluator side after candidate output is frozen, and fresh regeneration verifies every deterministic official per-review metric before emitting the counts. These results identify where the bounded probe fails; they do not establish superiority for the full system.

8.5 Failure analysis and external status

Under the evaluator's fixed terminal-failure precedence, the exploratory adaptive comparator exposes 296 premature-closure and 16 present-but-missed classifications while retaining 78 cases as `CANNOT_CHECK`; the no-dedup ablation exposes 193 budget-exhaustion failures. The governed system's only terminal failure classification is honest budget exhaustion on 59 tasks: 260 tasks pass, 71 are censored, and premature closure remains at zero. The full per-system taxonomy is archived beside the publication summary.

A gold-defined route-stop replay is archived as supplementary material. The governed system makes 13 route-stop false positives in 1,950 route-stop events (0.006667): those restricted-route cases still have a theoretically reachable gold identity when the route stops. These do not become task-level false closures. The unavailable-route obligation remains open, so the governed system leaves the task undetermined rather than asserting completeness. Among the 1,878 governed task-routes that reach the oracle exhaustion point, there are zero route-stop false negatives. The restricted route records 377 post-exhaustion attempts in total; no task takes the two or more further attempts required for a route-stop false negative. This null-on-FN result is reported rather than being silently omitted, and it demonstrates why route-stop error and task-stop safety must be measured separately.

The remaining external authorities are kept separate from the completed MetaSyn probe. AutoResearchBench Wide's released scorer is deterministic and credential-free, while the official Deep metric uses the released LLM title judge. That judge has now executed for the declared keyless public-arXiv probe: all 600 released Deep tasks were scored under the pinned judge prompt and model reached through a local protocol adapter, and the judge matched the reference title on 0 of them, a hit rate of 0.000. A nine-case positive/negative judge control—verbatim, truncated, and case-variant matches alongside distinct-paper non-matches—passed in full before the archive was accepted, so a zero-match outcome is attributable to the probe rather than to an always-no-match judge failure. This is a bounded external probe, not a matched Deep comparison against a baseline, and no Deep superiority claim is drawn from it. SAGE still lacks both the published retrieval corpus and an official evaluator. The frozen three-route live campaign also remains undetermined: its arXiv route is credential-free, but its two OpenAlex-backed route families require a funded OpenAlex API key, which the verified same-repository preflight did not find. No absent credential is converted into a negative outcome or a zero-cost measurement.

That zero was then diagnosed rather than left standing. A judge-free stage attribution over the same archive compared all 10,850 returned candidate titles against the 564 tasks that carried both a reference and candidates: exact-title recovery 0, substring recovery 0, and token-overlap recovery at a half-token threshold on 8 of them. Because a plain string matcher recovers almost nothing from the candidate sets themselves, the failing stage is candidate *generation* and not scoring, judging, or reference formatting. The attribution names a specific mechanism: questions whose incidental document-apparatus vocabulary—*supplementary*, *appendix*, *figure*—is treated by the acquisition mechanic as content vocabulary, so a title such as *Supplementary Orbit* outranks the paper the requested number actually sits in.

A successor acquisition mechanic was then built against that named mechanism and validated under a freeze written before any corpus was generated. The successor, a discriminative term-gating mechanic, uses index statistics only—never the gold set, the target, or the task family—discarding question terms whose document frequency exceeds five percent of the corpus, ranking the survivors by inverse document frequency at the baseline's own query width, and demoting rather than deleting documents that agree on fewer than two discriminative terms. On an authored corpus of 2,760 documents in which the named mechanism is present by construction and the denominator is complete, the current mechanic recovers the target for none of the 120 echo tasks within the top ten while the successor recovers all of them, at a rank-one recovery rate of 0.983. The paired McNemar test over those tasks is discordant on 120 pairs, every one of them favouring the successor. The harm guard is not vacuous: the current mechanic already scores 1.000 on the 60 tasks written without echo, and the successor does not damage them. BM25 over the same query recovers much of the top-ten rate (0.633) but almost none of the rank-one rate (0.0083), so the query-side gate rather than document-side weighting is what carries rank one. A pre-registered specificity control behaves as required: on 40 tasks whose question and target share no content token, the successor gains exactly 0.000, confirming the arena isolates the lexical mechanism instead of rewarding the successor generally.

This result is scoped to constructed reproduction only, and that scope was fixed before any number existed. The official probe cannot be re-run here: its frozen input is not in the repository and both public providers refuse this environment's proxy. The successor is therefore evidence that the mechanism the attribution names is by itself sufficient to drive a lexical candidate generator to near-zero recovery, and that a discriminativeness-grounded generator is not driven to zero by it. It licenses no statement about target-hit rates on AutoResearchBench Deep, does not revive the archived probe, and does not discharge the paper's successor-mechanic terminal, whose arena must be external and whose evaluator custody must sit outside the lane that authored the mechanic. One further limitation is declared in advance rather than discovered later: the successor requires corpus-level document frequencies, which are exact offline but against a live provider need a result-count endpoint or sampled index that this study does not supply.

A separate gold-blind OpenAIRE identity discriminator first established that structured arXiv persistent-identifier objects were available on 3 of 12 frozen public Wide questions, with 12 of

12 HTTP requests successful; that discriminator was forbidden to score or promote the paper. A subsequently frozen 400-row OpenAIRE/Crossref matched campaign then issued exactly 1,200 logical provider requests over one shared capture. The 400 Crossref discovery calls and 400 direct OpenAIRE calls succeeded, but every predeclared OpenAIRE DOI-crosswalk call returned HTTP 400. The observed provider-success fraction was therefore 0.666667, below the frozen 0.90 validity gate. The balanced baseline, direct-first diagnostic and governed projection were byte-identical, and their paired IoU differences were zero on all 399 distinct questions. The frozen analyzer correctly emitted an undetermined terminal, not a valid negative result.

Two further prospectively frozen campaigns followed, and together they correct the natural reading of that first one. A second campaign changed only the crosswalk encoding, from a single quoted OR expression to repeated persistent identifier parameters; its gold-blind transport probe returned HTTP 200 with an empty result array, below its prospectively frozen minimum of one, so acquisition never began and no score was produced. A third campaign changed the encoding again, to the documented identifier-filter OR list, and this time the transport probe validated: its receipt and raw response hashes were bound into the capture manifest and independently revalidated at score time. All 400 rows were acquired and scored. Provider validity reached 0.715833, again below the frozen 0.90, so the analyzer again emitted an undetermined terminal rather than a negative result.

The correction is this. The three systems were byte-identical in the first campaign, and their evaluator outputs were byte-identical again in the third, where the cross-backend route *was* valid and the three candidate files were genuinely distinct. So the identity is not explained by the failed crosswalk. It is explained by the measurement floor. The frozen decision rule requires an average-IoU difference of 0.03 before superiority is supported, and the best-scoring arm in both completed campaigns achieved an average IoU of 0.003924 and 0.004374 respectively — roughly a seventh of the difference the campaign exists to detect. Deleting a competing arm outright would not reach the threshold. Three systems that all retrieve almost nothing tie on nearly every question, so their aggregate scores coincide and their scored outputs are identical files.

That matters for how the accompanying interval should be read. Both campaigns report a paired bootstrap interval of $[0.0, 0.0]$ over 399 matched questions, with 399 ties and no wins or losses, from ten thousand resamples, at the same confidence level and percentile method the campaign froze. Taken at face value that is the strongest equivalence evidence a bootstrap can produce. It is not evidence of equivalence at all: resampling 399 identical differences returns a zero-width interval at any sample size, and the interval would appear exactly this conclusive with four questions rather than four hundred. We therefore report no equivalence or non-inferiority conclusion from either campaign. Three reporting defects are recorded with them: both artifacts publish the unseeded sampled maximum-IoU-at-k family that this paper's own scorer excludes by rule, that family decreases in k — which a maximum over the top k cannot do — and every runtime total is reported as exactly zero, which is untransported measurement rather than measured cost.

The terminals, workflow receipts and expiring Actions archives are retained in the repository, with checksummed mirrors beside them and a separate resolution audit; Section 14 names each location. A further campaign would require a new prospective freeze; none of these captures may be reinterpreted after the fact. The prerequisite is no longer a transport encoding. It is finding out why retrieval scores near zero on a benchmark whose gold is reachable, because a fourth transport version scored at the same floor would spend the same effort to learn the same nothing.

Therefore the externally supported discovery-benefit claim for the full system remains undetermined. The controlled-index result supports the narrower statement that the frozen governance mechanisms behave as designed under a complete denominator and matched host-owned budgets; the MetaSyn result supports only its declared external retrieval/screening probe. Bounded AutoResearchBench probe results, when archived, must remain separately labeled unless they exercise the final matched system under the frozen promotion plan.

9 Public cross-dataset screening: one-pool learner-rule transport

The acquisition ceiling separates route emission from downstream selection. We therefore tested one narrower transport question independently of open-world route invention: once a fixed citation pool has been acquired, does sequential label feedback retain value on a public source outside the SYNERGY review family used for controller development? The target was the CC BY 4.0 physiotherapy citation-screening corpus released by Dhrangadhariya, Hilfiker and Müller [5]. The pinned 53,881,052-byte file contains 25,540 tab-delimited raw lines.

Transport estimand. Here "transport" is component-specific. Across source families, the active update rule, its hyperparameters, and the batch-size rule were held fixed, but the record-identity adapter and TF–IDF representation were re-instantiated on the target corpus, and the initial seed was selected using target labels. Let P denote the canonical repaired pool and $z(P)$ that disclosed label-informed seed. The primary observed contrast is therefore

$$\Delta_{10}(P, z) = R_{\text{active}}(P, z; 0.10) - \max_{k \in \mathcal{K}_{\text{reg}}} R_k(P, z; 0.10),$$

conditional on this one P, this seed rule, and the frozen source-specific adapter, where $\mathcal{K}_{\text{reg}}$ contains the two registered passive within-pool baselines. It is not an estimator of a mean effect over review populations and does not transport a fitted model, representation, identity layer, initialization policy, or cold-start capability.

Schema incident and successor identity. The source has no header. A preflight intended to inspect a header therefore exposed the first row's public label before the protocol was frozen. That exact raw line was excluded by SHA-256, and the campaign is explicitly not outcome-blind. V1 then failed its binding gate before any model outcome: two of the remaining rows used noncanonical class literals and three canonical rows collided with earlier rows under the frozen class-free record identity. The V1 transport binding therefore remains undetermined rather than resolved.

Before computing any model score, V2 froze the schema repair: admit exact `include`/`exclude` rows only and retain the earliest canonical row within an identity collision. The resulting population contained 25,534 unique records, including 2,259 inclusions. This repair makes V2 post-schema-development evidence; it does not restore outcome blindness or independent custody.

Frozen arms and endpoints. All arms used the same label-informed warm start: the complete public labels were scanned to hash-select one known inclusion and one known exclusion. Results are therefore conditional on an available positive seed and do not measure cold-start screening. A passive static comparator ranked the remaining L2-normalized title–abstract–MeSH TF–IDF vectors by cosine similarity to the included seed; a random comparator averaged 20 frozen permutations. The active arm refit a class-balanced logistic-loss SGD classifier after batches of 52 newly screened records and selected the highest predicted inclusion probabilities. The primary endpoint was recall after screening 10% of the population. Secondary endpoints were recall at 5% and 20%, fraction screened to 95% recall, and WSS@95 $= 0.95 - f_{95}$. V2 required exact binding, active recall at 10% of at least 0.50, advantage of at least 0.10 over the better-performing registered baseline, positive WSS@95, and no harm relative to random.

Result. All five declared gates passed in the producing lane; the receipt records matching protocol and implementation hashes. At 10% screened, active recall was 0.703409 versus 0.335989 for the better-performing of the two registered passive within-pool baselines, the static seed centroid, a difference of +0.367419; the 20-permutation random mean was 0.102789. Active recall was 0.406375 at 5% and 0.945994 at 20%. It reached 95% recall after screening 0.205256 of the pool, giving WSS@95 of 0.744744; the static comparator's corresponding values were 0.676941 and 0.273059. The producing lane reports a byte-identical local rerun, but no separate replay receipt or independent verification is archived.

Arm	R@5%	R@10%	R@20%	f_{95}	WSS@95
Active label feedback	0.406375	0.703409	0.945994	0.205256	0.744744
Static seed centroid	0.177512	0.335989	0.560425	0.676941	0.273059
Random, 20-seed mean	0.051439	0.102789	0.201239	0.950305	-0.000305

Table 2: Frozen one-pool public screening transport result. Recall columns report the included-record fraction recovered at each screening budget; f_{95} is the population fraction screened to reach 95% recall. Random is the arithmetic mean over 20 frozen permutations. The analysis unit is one post-schema-repair pool, so the table carries no cross-world inferential claim.

Post-V2 active-comparator audit. Because both original comparators were passive, the +0.367419 contrast does not identify superiority over a modern active-screening controller. After the V2 outcome was known but before either new comparator outcome was opened, a distinct audit froze the pinned ASReview ELAS u4 and u3 model components [15]. Both cadence-matched arms received the same repaired record universe, text fields, label-informed warm seed and batches of 52 labels as the candidate. This reproduces the pinned model, balancing, TF–IDF and maximum-query components under a matched feedback cadence; it is not an exact execution of the ASReview application or its default one-record query cadence.

The V2 candidate reproduced exactly. ELAS u4 was nevertheless stronger: recall at 10% was 0.722444 versus 0.703409, so the candidate-minus-u4 difference was −0.019035. U4 also achieved WSS@95 of 0.746781 versus 0.744744, a candidate difference of −0.002037. U3 was weaker than the candidate, with recall at 10% of 0.667109 and WSS@95 of 0.713296. Thus the prefrozen +0.05 primary-margin and nonnegative WSS-dominance gates both failed, so the registered active comparator ties or wins and a controller successor is required. The original V1 failure and V2 one-pool terminal remain immutable; this audit changes the admissible interpretation, not their identities.

The resulting positive content is narrower but more diagnostic: sequential feedback substantially outperformed the two passive arms on this pool, while the particular logistic controller did not exceed the stronger active family. No post-outcome tuning on this pool can promote the claim. A controller-level superiority claim now requires a prospectively frozen, source-disjoint multi-review family whose comparator registry includes the u4 family before protected outcomes are opened.

This is a large conditional within-pool contrast after fitting a new active SGD model with the previously fixed estimator family, hyperparameters and batch-size rule, but a source-specific title–abstract–MeSH representation and identity adapter. It is not an inferential replication across review worlds: the analysis unit is one pool, so no cross-world p-value is reported. Nor does it identify an ORION-specific effect, because active screening is a donor component and the fixed pool contains no open-world route invention or scientific-closure decision. It is also conditional on the label-informed warm start and does not establish cold-start transport. Publication-grade promotion still requires a prospectively population-frozen, source-disjoint multi-review family with protected labels, the now-identified stronger active comparator family, and external custody.

Cross-review SWIFT successor. A later public-development successor tested the fixed candidate against the cadence-matched pinned u4 components across five SWIFT review decisions released with the SWIFT-Review workbench [8]. V1 and V2 stopped before fitting either model and remain CANNOT-CHECK mechanics failures: V1 used a casefolded newline-delimited identity that did not match the preflight counts, while V2 exposed that the original direct-review preflight had inserted spaces between characters. V3 instead froze the intended case-sensitive whitespace-normalized, label-independent identity audit. That correction occurred after review class counts were opened but before any candidate or comparator outcome, so V3 is public-development evidence rather than protected confirmation.

The corrected population contained 96,241 canonical rows across BPA, Fluoride, Neuropain, PFOS–PFOA and Transgenerational. The unweighted review-level mean candidate-minus-u4 recall@10% was -0.021717971642, against a preregistered $+0.05$ primary margin; mean candidate-minus-u4 WSS@95 was -0.050396333672, against a nonnegative relative-work-saving gate. U4 was better on both endpoints in all five reviews. The worst recall@10% difference was -0.039215686275 for Fluoride, so the -0.05 harm gate passed, and candidate WSS@95 was positive in every review, so the absolute-work-saving gate passed. Those two gates are non-compensatory and do not repair failure of the primary and relative-work-saving gates.

Gate or estimand	Criterion	V3 result
Mean candidate-minus-u4 R@10%	$\geq +0.05$	-0.021717971642 (fail)
Mean candidate-minus-u4 WSS@95	≥ 0	-0.050396333672 (fail)
Worst-review R@10% difference	≥ -0.05	-0.039215686275 (pass)
Candidate WSS@95 in every review	> 0	pass

Table 3: SWIFT V3 cross-review public-development decision. Means are unweighted over five distinct review decisions, not over 96,241 pooled rows. The reviews are not asserted to be statistically independent; the maximum pairwise shared-content count was 68. Passing the harm and absolute-work-saving gates cannot compensate for the failed primary and relative gates.

V3 therefore records the controller as failing one or more public development gates across the cross-review panel, and requires a successor. The earlier one-pool outcome stands unchanged. The result is not cold-start evidence, exact ASReview application execution, protected confirmation, independent custody, population transport, ASReview software inferiority, or ORION-specific superiority.

V4 executed the frozen 2-by-2 representation-by-learner/balancer factorization only as post-outcome mechanism diagnosis. All four arms used the same 96,241 canonical rows per arm, and the candidate and u4 corner reproductions matched V3 exactly. The unweighted review-level learner/balancer main effect was $+0.022618035379$ for recall@10% with the aggregate sign in five of five reviews, and $+0.041725819160$ for WSS@95. The representation main effect on recall@10% was -0.000900063738, with its aggregate sign in only one of five reviews; the interaction was $+0.003067576381$, with its aggregate sign in three of five. Under the frozen classification, only the learner/balancer component was stable on this panel.

V4 therefore records a stable component under post-outcome factorization, and requires a content-disjoint family whose outcomes have not been opened before that component can be used. Because the factorization was designed after the V3 outcomes and reused the same partly overlapping decisions, it cannot replace the candidate, promote any arm, or support a superiority claim on SWIFT. The selected mechanism must be frozen without outcome access and evaluated on a content-disjoint review family with the same active u4 comparator and unchanged safety gates.

V5 performed that test under a new identity on the CC0 SYNERGY V1 release. Before label values or model outcomes were opened, the study selected the five largest review decisions from publisher-documented counts and froze the population, four-arm implementation and unchanged gates. A label-blind adapter excluded every normalized content identity appearing in SWIFT and every identity shared across the five successor reviews. The retained units were the five review decisions, not the 101,044 screening records: Walker (40,442), Brouwer (38,009), Hall (8,783), Wassenaar (6,621), and Leenaars (7,189). Final SWIFT overlap and pairwise successor overlap were both zero.

The learner/balancer recall@10% main effect was +0.008834277869, below the locked +0.010858985821 magnitude threshold. It was strictly positive in three of five decisions and exactly zero in Brouwer and Hall, failing the four-of-five strictly-positive sign gate. The same contrast was positive in the frozen recall@5%, recall@20%, and WSS@95 diagnostics (+0.031876154087, +0.019244987149, and +0.030373086810), but those diagnostics do not replace the failed recall@10% rule. The preserved full-arm candidate-minus-u4 means were −0.011841648130 at recall@10% and −0.041523905587 at WSS@95, so the original relative margin and work-saving gates again failed while harm and absolute work saving passed without compensation.

V5 therefore records the content-disjoint attribution as failing one or more frozen transport gates, and requires a successor. This is a preregistered adverse same-session transport result, not independent confirmation. All five decisions share one release, representation and local adapter, and their labels encode final full-text inclusion rather than an application-exact title/abstract decision. No arm is promoted. A successor must use a lawful family disjoint from both SWIFT and V5, keep recall@10% as a locked adverse coprimary endpoint, and preregister a continuous recall–effort curve before outcomes, ideally under independent custody.

9.1 V6 outcome-unopened source and continuous-estimand freeze

V6 closes the next source-feasibility discriminator without executing a comparison. It binds all 14 revision-one CSV review decisions exposed by the public OSF node `vt3n4`, Medical Guidelines Dutch Association Medical Specialists. The node declares CC BY 4.0; every file is fixed by OSF file and GUID identity, revision-one URL, byte count and SHA-256. Label-blind canonicalization used only stable key, title, abstract and the declared PubMed field. It reduced 5,074 raw rows to 4,934 canonical rows. One provisional row matched a V5 raw-work content identity, and 65 identities occurred in more than one KIFMS review, affecting 132 rows; all were removed symmetrically before any KIFMS label or model outcome. Final normalized title–abstract content overlap with both SWIFT and V5 is zero. The declared PubMed field is empty in all 5,074 rows, so zero PMID overlap is vacuous rather than a second corroborating channel. The final content-union digest is recorded in the availability statement.

For a complete ordering $i = 1, \ldots, N_r$, hidden binary labels y_i and $P_r = \sum_i y_i$, the new coprimary functional is frozen as

$$\mathrm{CRE20}_r = \frac{1}{0.20 P_r} \sum_{i=1}^{N_r} y_i \max\left(0.20 - \frac{i}{N_r}, 0\right),$$

the normalized area under the complete recall–effort step function over the first 20% screened. Recall@10% remains an adverse coprimary endpoint under the unchanged magnitude threshold;

both endpoints require a strictly positive effect in at least 12 of 14 decisions. The active u4 comparator, full factorial cells, batch and seed rules, learner and full-arm WSS@95 gates, worst-review harm gates and absolute-work-saving gate are unchanged. No endpoint can compensate another.

This freeze opened no KIFMS label value, class count, seed, V6 ordering or comparative score. It is outcome-unopened only for the exact V6 design context: the companion repository contains paths to historical public ASReview simulation artifacts for the named datasets, whose metric contents were not inspected here. The family is therefore not described as globally unstudied or historically outcome-sealed. Public source-body feasibility is established, but the selected `expert_inclusion` semantics, two-class existence in every review, a signed runner, and independent protected custody remain unresolved. V6 therefore records a lawful, exact, source- and label-blind disjoint population as frozen, with independent protected execution still undetermined. It supplies no performance, harm, work-saving, transport or superiority result.

9.2 V7 hash-frozen transparent public execution

V7 separates an unavailable strengthening condition from the empirical critical path. Before opening any KIFMS label value, it froze a new identity, the exact V6 source/population hashes, the four-arm runner, dependency versions, CRE20/R@10 coprimary rules and every unchanged harm/work-saving gate. The 14 revision-one CC-BY-4.0 files then matched their exact bytes and SHA-256 values; an outcome-blind unique single-exclusion reconstruction reproduced all 4,934 canonical rows and the V6 union hash. Every review contained both `expert_inclusion` classes, and every arm produced a complete order.

The learner/balancer main effect is positive on average: +0.084471534 on CRE20, +0.148820475 at recall@10%, and +0.119813186 on WSS@95. Thus both magnitude gates and mean work saving pass. CRE20 is strictly positive in 11/14 reviews and recall@10% in 9/14, below the locked 12/14 sign gates. The exact descriptive two-sided sign-test values are 0.057373 and 0.423950. The worst-review learner recall@10% effect is −0.125, failing the −0.05 harm floor. The full candidate-minus-u4 means are −0.201647856 at recall@10% and −0.119159673 on WSS@95, so the relative margin, work-saving and harm gates also fail; candidate absolute WSS95 remains positive in every review.

V7 therefore records the transparent public execution as failing one or more locked performance gates, and requires a successor. This is an adverse result under six of ten locked performance gates. It is also a sharper upward diagnosis: the u4 learner/balancer is the productive component across both representations, while the standalone candidate and complete candidate stack do not envelop the strong donor. The next mechanism therefore makes u4 mandatory, learns only a leave-one-review-out residual correction with a hard u4 fallback, and requires a source-disjoint public family for its next test. V7 is same-workspace transparent public development, explicitly records independent custody as false, and cannot issue confirmation, ASReview-software or domain-general superiority.

10 Post-saturation unresolved-route authority

The post-saturation literature search raises the comparison standard rather than shrinking the systems question. Verification-aware claim coverage, decision-sufficiency and utility stopping, coverage-driven deep-research completion, finite source-obligation certificates, and generic workflow-obligation graphs are all granted as donor mechanisms [13, 2, 4, 6, 21]. P2-X then asks the remaining scientific question: after those mechanisms have extracted everything they can from the routes that are actually available, what authority exists to close the task when a *material* route is unavailable, censored, provider-invalid, or still carries an unresolved question-conditioned processing obligation? The post-saturation literature search made a wider systems claim testable only after granting more prior art to the acquisition layer. Verification-aware claim coverage, decision-sufficiency and utility stopping, coverage-driven deep-research completion, finite source-obligation certificates, and generic workflow-obligation graphs are all treated as donor mechanisms rather than as novelty claimed here [13, 2, 4, 6, 21]. The remaining question is narrower: after those mechanisms have done everything they can on the routes that are actually available, what scientific authority exists to close the task when a *material* route is unavailable, censored, provider-invalid, or still carries an unresolved question-conditioned processing obligation?

The exact-contract battery. The battery freezes five acquisition families—scientific literature, web/API evidence, dataset registries, code/artifact repositories, and experiment/measurement routes—with eight hostile archetypes per family. The cases distinguish all-obligations-discharged closure, one available route remaining open, unavailable/censored/provider-invalid material routes, duplicate route coverage, already-acquired but not question-processed content, and low-utility hard routes that stay open. Route-local decisions are stop, continue, or undetermined; task terminals are close, continue, or undetermined, with undetermined reported as its own outcome rather than collapsed into a refusal to close. Route order is seed-permuted, and held-out variants add unavailable *non-material* decoy routes so that blanket "any unavailable source blocks" logic cannot pass.

Strong donor products. The primary B1 comparator is deliberately donor-complete. It receives route identity/materiality, HALT-like evidence coverage, decision sufficiency, utility stopping, finite route certificates, deduplication, route state, and the question-conditioned processing coordinate. It also makes the same correct route-local decisions as P2-X. Its only systematic difference is global aggregation: once all *available* material routes locally stop and their acquired content is processed, it treats the task as closed, excluding unavailable/censored/provider-invalid material routes from the closure denominator. B2 allows a global sufficiency signal to authorize task stop. B3 is an ideal information-equivalent product given the exact P2-X unresolved-route semantics and is expected to tie; it tests portability of the authority rule rather than serving as a weak baseline.

Protected result: exact closure advantage. On 400 prospectively frozen protected cases, P2-X obtains **400/400 exact scientific closure decisions** (ESCD = 1.000), versus 250/400 (0.625) for the donor-complete B1 product and 100/400 (0.250) for B2. The paired P2-X–B1 advantage is +0.375 with a predeclared domain-stratified bootstrap 95% interval $[0.3275, 0.4225]$, above the registered +0.10 practical margin; exact McNemar discordance is $b = 150, c = 0$. P2-X records **zero false task closures**, preserves 1.000 clean task-stop accuracy when every material obligation is discharged, and has a positive advantage over B1 in all five acquisition families. B1's 150 ESCD errors are exactly the three unresolved-material-route archetypes—unavailable, censored, and provider-invalid—while it passes every other archetype.

Strong products. The primary comparator, B1, is deliberately donor-complete. It receives route identity/materiality, HALT-like evidence coverage, decision sufficiency, utility stopping, finite route certificates, deduplication, route state, and the question-conditioned processing coordinate. It also makes the same correct route-local decisions as the authority rule under test. Its only systematic difference is global aggregation: once all *available* material routes locally stop and their acquired content is processed, it treats the task as closed, excluding unavailable/censored/provider-invalid material routes from the closure denominator. A second comparator, B2, allows a global sufficiency signal to authorize task stop. A third, B3, is an ideal information-equivalent product given the exact unresolved-route semantics of the authority rule and is expected to tie; it is an expressivity boundary rather than a weak baseline.

Protected result. On 400 prospectively frozen protected cases, the authority rule obtains 400/400 exact scientific closure decisions (ESCD = 1.000), versus 250/400 (0.625) for B1 and 100/400 (0.250) for B2. The paired advantage over B1 is +0.375 with a predeclared domain-stratified bootstrap 95% interval $[0.3275, 0.4225]$, above the registered +0.10 practical margin; exact McNemar counts are $b = 150, c = 0$. The authority rule records zero false task closures, preserves 1.000 clean task-stop accuracy when every material obligation is discharged, and has a positive advantage over B1 in all five acquisition families. B1's 150 closure failures are exactly the three unresolved-material-route archetypes—unavailable, censored, and provider-invalid—while it passes every other archetype.

An independent implementation reproduces all 400 case identities, protected-bundle and canonical decision-row hashes, every headline arm count, and zero B3 decision mismatches. The non-material unavailable decoys cause no materiality error, and all statistics include the held-out decoy variants.

Portability result. B3 also obtains 400/400 and is decision-identical to P2-X. This is a positive implementation result: the unresolved-route authority rule is sufficient and portable. It can be realized as the ORION closure layer or as a correctly enriched donor product without changing any registered decision. The P2-X advantage over B1 therefore isolates a specific missing coupling in a realistic donor-complete interface rather than depending on centralized expressivity or on weak stopping machinery.

Adverse external probes as failure-localization and transport-authority evidence. The retained external probes strengthen the closure interpretation when read at their actual authority rather than treated as failed attempts at the P2-X headline. On the 86-review MetaSyn ID-only probe, retrieval recall is 0.7485 while final inclusion recall is 0.5651; conditional retention is 0.7224 and measured post-retrieval loss is 0.1834. This directly localizes a substantial loss-bearing stage after retrieval and shows why acquisition success and question-conditioned processing completion should not be collapsed into one stop signal. On the 600-task Deep keyless public-arXiv stress probe, the official title judge records 0/600 hits after a 9/9 judge-control pass. This does not estimate full ORION performance; it is an extreme route-inadequacy example in which a cleanly executed accessible route plainly cannot authorize task closure.

The prospective OpenAIRE/Crossref Wide campaign provides a complementary evaluation-level authority result. Candidate contracts, hashes, common capture, request count, and candidate cap all validate, yet logical provider validity is only 0.666667 and the transport-validity gate fails. The scientific terminal is therefore `P2_WIDE_EXTERNAL_CANNOT_CHECK`: the framework refuses to mint either superiority or a valid null from a non-transportable comparison. This is the same fail-closed principle at the evaluation layer. A locally well-formed run is not automatically an authority to close the scientific comparison.

The widest supported conclusion is therefore larger than generic search stopping. **Scientific closure is a governed proof obligation over material acquisition and processing routes, and evaluation transport is itself an authority-bearing route.** Route-local success, a completed API call, a successful retrieval stage, or a mechanically produced benchmark score cannot self-authorize task-global completion when a material obligation remains unresolved. P2-X provides the protected 400/400 decision result; the adverse external probes independently show why stage identity and transport validity have to remain explicit rather than being hidden inside a single success signal.

Boundary result. B3 obtains 400/400 and is decision-identical to the authority rule. Thus the result does not establish inherent expressivity, centralization superiority, or ownership of stopping/obligation machinery. It supports a bounded authority distinction: with realistic donor-complete interfaces, an unavailable/censored/provider-invalid *material* acquisition route must stay unresolved rather than disappearing from the task-closure denominator. An ideal product supplied that same rule is equivalent.

The result also does not repair the historical external Wide campaign. The OpenAIRE/Crossref campaign is still provider-invalid and undetermined; no independently implemented retrieval engines or new live multi-provider campaign were executed in the battery. Deployed and retrieval-engine generality therefore stays undetermined.

11 Successor interface: structure-conditioned and source-grounded acquisition

The acquisition–closure separation also applies when a search route is generated from a structural need rather than topical vocabulary. A successor route may derive its query from a method signature, a failure signature, a representation analogy, or an unresolved atom study. This remains an acquisition choice. The label "structural" does not make a route independent: backend identity, query-derivation identity and capture identity must still earn independence under the accounting used elsewhere in this paper.

Returned material is retained as a candidate donor, not a transfer certificate. Before a structurally interesting candidate is passed downstream, its assumptions, protected invariants, effects and reconstruction obligations can expose an acquisition-local obstruction or unresolved mapping. Semantic distance is not evidence of novelty, and surface similarity is not evidence of valid transfer. Whether a correspondence actually holds, and whether a candidate deserves scientific promotion, are separate problems that this paper does not decide.

Two source-grounded historical substrates are available to this acquisition layer, with intentionally limited authority. A discovery morphology atlas freezes a 51-episode, four-domain, ten-coordinate descriptive basis with explicit no-Jump controls and two post-basis no-material-change rounds. A failure atlas freezes a 30-episode revision-aware schema. In both cases, historical material remains mechanism-extraction and benchmark-design evidence: private cognitive steps stay non-inferable, competing interpretations are retained, and curation does not transform famous examples into protected confirmatory gold. The stronger corpus-ready state of the failure atlas was not claimed because independent domain-expert agreement was not obtained.

The prospective atom results reinforce the same boundary. Broad tension, imagination and concept labels decomposed into multiple candidate atoms rather than becoming new search-route authorities, and the zero-error Jump candidate tied the verified representation-regime parent on both frozen splits. None of those outcomes authorizes an unresolved route to be closed, corpus completeness to be inferred, or a historical analogy to be promoted into scientific evidence.

Route-local stopping therefore remains local under this extension. Exhaustion, censorship, provider failure, budget stopping, a temporarily empty structural route, or a stable descriptive atlas does not establish that no useful source or method exists. A route stays open or revisitable whenever a new representation, source family, or upstream structural object could materially change the search. The successor interface extends the ways candidates can be acquired without weakening the central rule: acquisition evidence does not acquire scientific-closure authority merely because it came from a richer representation or a curated scientific history.

12 Related work

AutoResearchBench establishes that both specific-target and open-ended scientific-literature discovery remain difficult for current agents [19]. The acquisition frontier is already strong and heterogeneous: SAGE shows that tuned lexical retrieval can outperform LLM retrievers inside deep-research workflows [9]; AgentIR uses the agent's reasoning trace as a retrieval signal [3]; SIEVE combines fielded Boolean admission, structure-rich inspection and selective section fetching [17]; and breadth-first bibliography expansion recovers material missed by API-only retrieval [14]. Rather than reserve a weaker baseline, Definition 1 places all runnable instances and their outcome-blind functional analogues inside the donor-complete policy class. Their actions enlarge the acquisition envelope; a candidate contribution must survive that enlargement.

Screening and stage attribution form a second neighbourhood. MetaSyn separates retrieval from eligibility screening and shows why a retrieval ceiling does not imply an inclusion-recall ceiling [18], and AgentSLR shows that end-to-end systematic-review automation is an active baseline family in its own right [12]. This paper adopts their measurement discipline—stage attribution and denominator caution—and inherits their warning that one aggregate recall number hides where the evidence was actually lost.

The registered external retrieval comparison uses the TREC-COVID collection [16]; its adverse result is reported as bounded transport evidence rather than generalized to open-world discovery.

The stopping and obligation literature owns more than "when to stop." DeepControl and decision-theoretic screening optimize continuation against information value, cost and payoff [20, 7]; HALT uses claim–evidence coverage and abstains when it cannot verify that coverage [13]; MiCP controls adaptive coverage [22]; and structured Search-R1 learns sufficiency and gap judgements [10]. Don't Stop Early couples coverage-driven objectives to evidence-aware completion [4], while Confidence-Based Stopping targets decision sufficiency rather than recall alone [6]. KnowPlan provides finite source-obligation certificates [2]; iCORE binds work and obligation transitions to evidence [21]; and Science of Intent explicitly studies closure gaps and misclosure [1]. The present theory grants all of these mechanisms to the donor state. Its residual is the factorization question in Theorem 2: after composition, does the retained state actually distinguish every closable world from every materially open world? If yes, the product is sufficient and may tie. If not, no new classifier over the same state can manufacture authority.

Question-conditioned memory has its own line of work: MemChain constructs question-conditioned evidence plans and grounded memory traces [11]. The envelope treats memory as an optionality-bearing scientific state, not merely a token cache. Content identity prevents duplicate rediscovery from inflating coverage, whereas question-conditioned processing state permits a changed extraction frame to reopen already encountered material. Whether that additional coordinate matters is decided by fibre separation and prospectively hidden future questions, not by claiming generic memory novelty.

13 Limitations and integrity

The completed controlled experiment is deliberately synthetic. Its complete denominator is what makes missed relevant material and premature closure observable, but the authored routes, public probe vocabulary and easy screening text make it a mechanism test rather than a realistic estimate of retrieval difficulty. The observed +0.31282 recall difference therefore cannot be transferred to the open web or to AutoResearchBench by analogy.

The contribution is a comparison and composition theory rather than a claim to own its components. Scientific retrieval-augmented generation, agentic literature search, automated systematic review, capture–recapture estimation, lexical, dense, reasoning-aware, field-aware and citation-expansion retrieval, learned or decision-theoretic stopping, conformal coverage, finite obligation ledgers and question-conditioned memory are prior art and enter the donor class. The claimed theoretical residual is the joint acquisition–authority envelope, its run-conditional acquisition ceiling, the closure-factorization criterion, the indistinguishable-world lower bound, the maximally permissive robust safe set, and the completeness condition for an explicit obligation rule.

The suite carries 390 tasks, which reaches the plan's committed precision tier at an achieved half-width of 0.0496 for a standalone rate. The frozen statistics plan fixed those tiers before outcomes were inspected and assigns this campaign that committed authority; because the achieved half-width still exceeds the frozen superiority margin of 0.03, the primary offline result carries the plan's underpowered label whatever the task count becomes. All 16,380 system/task/repeat executions reproduce exactly, and the 3 deterministic repeat seeds demonstrate stable execution; repeats nest within task and never increase the statistical unit.

The exact-contract battery broadens the *contract families* tested, not the realism of retrieval. Its 400 cases span five acquisition settings and include seed-permuted route order plus non-material unavailable decoys, but route state, route materiality and question-processing state are explicitly supplied rather than inferred from noisy web behavior. The 1.000 exact-closure ceiling therefore supports the fail-closed authority rule only on the registered exact contracts. No independently implemented lexical/dense/reasoning retrievers or new live provider campaign were used to establish retrieval-engine generality, and the ideal-product arm shows that an ideal product given identical unresolved-route semantics is extensionally equivalent.

Open-world completeness remains generally unprovable from finite search. External providers are mutable, rate limited and differently scoped, and public benchmark questions and gold identifiers create search-time contamination risk. The corrected MetaSyn run is external but narrow: it uses pinned BM25 retrieval plus a deterministic public-protocol screening rule and the official ID-only evaluator, not the complete route system. Its exact evaluator-side ledger contains 1,677 reference links, of which 598 are missed at retrieval and another 343 are lost after retrieval, leaving 941 final false negatives. These numbers characterize that probe and are not promoted to a general claim about the system.

The Zenodo transport result is also deliberately narrower than the envelope claim. Its labels are public, its first row was exposed before protocol freeze, and V2's eligible population was frozen only after V1 revealed two noncanonical class rows and three duplicate identities. The active estimator family, hyperparameters and batch-size rule were frozen before V2 scoring and outperformed the registered passive comparators after re-instantiation on the source-specific representation and identity adapter, but the analysis contains one physiotherapy pool, no independent label custody, no application-exact source-native screening reproduction, no open-world route action and no task closure. In the separate post-V2 active-comparator audit, cadence-matched pinned ELAS u4 reached recall 0.722444 versus 0.703409 and WSS@95 0.746781 versus 0.744744, failing the candidate's strict-margin and dominance gates. The evidence therefore supports a large conditional within-pool contrast against passive screening under a disclosed post-schema repair and label-informed warm start, not active-controller superiority, fitted-model, representation, initialization, cold-start, system-level, population-average, or cross-domain transport.

The later SWIFT cross-review successor broadens the number of public screening decisions but remains development evidence. V1 and V2 stopped before fitting either model because their population-identity mechanics did not bind the intended preflight population. V3 corrected that audit before any candidate/comparator outcome but after class counts were opened. Across five reviews and 96,241 canonical rows, the candidate-minus-u4 unweighted review-level means were -0.021717971642 for recall at 10% screened and -0.050396333672 for WSS@95; u4 was better on both endpoints in all five reviews. The primary-margin and relative-work-saving gates failed. Harm and absolute work saving passed but are non-compensatory. The reviews are distinct screening decisions, not proven independent samples (maximum pairwise shared content count 68), and the execution is not cold-start evidence, an exact ASReview application run, protected confirmation, independent custody, population transport, ASReview software inferiority, or ORION-specific superiority. The V4 component factorization was frozen only after V3 outcomes and is diagnosis only; it cannot promote a controller on SWIFT.

The theory does not remove that impossibility. The maximality theorem is relative to a declared compatible-world class; it cannot prove that a finite open-web route registry contains every material route or that the registered obligation rule recognizes every robustly safe history. The acquisition ceiling is conditional on the captured run and on a valid denominator, not an estimate of the best route that might be invented later. The envelope itself depends on the task distribution, budget, access interface and donor class, so cross-domain superiority is an empirical hypothesis rather than a corollary of the formalism.

The V12–V15 source-state sequence does not supply that missing empirical evidence. V12's keyword difference is post-gate diagnosis because its content identity diverged from V10. V13 establishes a genuine fibre-separating metadata coordinate, but only four of seven reviews support it, below the unchanged six-of-seven sign gates. V14 localizes a frozen commit–path mismatch and stops before any census or performance operation. V15 binds a coherent signed snapshot with 61 dataset blobs, but its exact root licence is MIT and must not be blended with current CC0 repository metadata. Its frozen candidate template also omitted V14's `/output/` path component; that template was never executed, and the V15B correction is authorized only for a successor. Exact index attestation returned 404, all three independent custody roles remain open, and no population or performance conclusion is available.

Several external authorities remain partially unavailable rather than negative. AutoResearch-Bench Deep official scoring has now executed through a local OpenAI-compatible adapter for the bounded keyless probe, with the judge control passing in full, but the matched multi-provider Deep comparison remains unexecuted. SAGE's paper-linked repository is data-only at the audited revision and supplies neither an official runner nor a licence; its published retrieval setting therefore remains unavailable as an arm. The separately archived OpenAIRE/Crossref Wide campaign failed its prospectively frozen provider-validity gate, so the matched external superiority claim remains undetermined; its observed zero paired effect is not a valid null or negative result. The exact-contract battery does not repair or supersede that external failure. Provider or credential unavailability is likewise kept undetermined, never converted into evidence of absence or a flattering zero-cost result.

The comparator-binding audit does identify SIEVE/SkimSearchAgent as a credible Apache-2.0 external acquisition candidate at the audited revision recorded in the availability statement. A one-instance keyless fixture completed with a stub policy and emitted the expected trace, rows and result artifacts. That is entrypoint conformance only: the zero fixture score is not a scientific result, and native answer/end/error/timeout trajectories do not authorize P2 closure. The outcome-blind adapter preflight forbids every such conversion, and the recorded outcome is that the fixture runs while explicitly not becoming a frozen comparator arm. Archived task-world reproduction and equal provider, route, budget, model, terminal and scoring bindings remain mandatory before

SIEVE becomes a comparator arm.

The 768-cluster successor also remains entirely prospective. Its arithmetic freeze establishes neither that the task worlds exist in compliant form nor that the planned effect is real. Four source-disjoint arena snapshots, reproduced baseline worlds, a capability-matched external comparator, live response/cost receipts and independently separated evaluator/gold custody are still missing. No part of that protocol is counted as external evidence in the present paper.

The mechanism ablations themselves impose an interpretive limit. Removing the question coordinate from the read ledger changes legitimate reread behavior but does not change aggregate recall in this suite; removing the unavailable-route open state changes censoring/closure semantics while leaving aggregate recall numerically unchanged. These null-on-recall mechanism findings are retained rather than converted into a stronger claim.

All result-bearing statements in the abstract, Results and conclusion are mapped to immutable archived artifacts through the claim ledger, the current non-promotion ledger, and the exact-contract result and verification receipts; the artifact names and paths are listed in Data and code availability. Rows without an archived supporting result stay undetermined; an access audit, implementation test, or credential preflight cannot substitute for a system outcome.

14 Data and code availability

The implementation repository contains the controlled index, host-owned evaluation harness, the frozen offline systems and ablations, the descriptive analysis code, and the publication-layer regeneration commands. Every artifact named in the narrative is listed here with its path, so that no repository location has to be carried in the prose.

Preregistration

The study was preregistered and frozen before any outcome was accessed, under the identifier `P2.open-world-discovery.v1`. A reader checking whether a reported analysis was planned or chosen after the fact should compare against that freeze; it is the reason every number below can be traced to a decision made in advance of seeing it.

Frozen design and measurement rules

The design freeze is `protocol/PROTOCOL_V1.json`; the measurement and statistical rules are `protocol/MEASUREMENT_PLAN_V1.md` and `protocol/STATISTICAL_PLAN_V1.json`; external-artifact access states are `protocol/EXTERNAL_ACCESS_AUDIT_V1.json`; and the prospective execution binding for the result-bearing offline campaign, including the exact subject revision and Git blob identities, is `protocol/OFFLINE_RUN_MANIFEST_V1.json`, which also binds the frozen identifier of every system under test. The identifiers are `orion_full` for the governed system; `bm25_keyword`, `dense_retrieval`, `sparse_dense_hybrid`, `one_pass_rag`, `agentic_single_route` and `protocol_driven_systematic_review` for the six confirmatory comparators; and `adaptive_multiroute_exploratory` for the explicitly exploratory one. Each is a key of the systems map in `evidence/offline_results/RESULTS_SUMMARY_V1.json`, alongside the six ablation identifiers. The mapping from every result-bearing sentence in this manuscript to the archived artifact key that supports it is `protocol/CLAIM_LEDGER_V1.json`, with a human-readable view at `evidence/CLAIM_LEDGER_V1.md`.

The definitions in Section 5 are implemented in `src/orion/knowledge/`: work identity and read state in `identity.py` and `ledger.py`, route independence and coverage in `routes.py`, and route control in `route_control.py`.

Offline complete-gold companion

The frozen complete-gold world is distributed under `evidence/offline_gold/`; it is fully synthetic, regenerable from the recorded seed, and verified on every clean CI run against its committed content hashes and suite fingerprint.

The compact publication result is `evidence/offline_results/RESULTS_SUMMARY_V1.json`, and the mechanism projection behind Figures 4–5 is `evidence/offline_results/OFFLINE_MECHANISMS_V1.json`. The full per-system failure taxonomy is `evidence/offline_results/TABLE_P2-3_failure_taxonomy.md` and the gold-defined route-stop replay is `evidence/offline_results/TABLE_P2-S1_route_stop_oracle.md`. The command `python3 papers/orion-12-open-world-scientific-discovery/scripts/ run_offline_companion.py --check` reconstructs all 16,380 runs and refuses drift. With `--write-raw DIR` it also emits the complete normalized JSONL result set and all rich run artifacts. The committed summary records SHA-256 digests of both complete sets, so a reproduction can be checked without requiring hundreds of generated files in the repository.

External benchmark artifacts

External benchmark artifacts are not redistributed. Their licences, access states and pinned revisions are recorded per artifact in `protocol/EXTERNAL_ACCESS_AUDIT_V1.json`, with source checks retained under `evidence/access/` and summarized in `protocol/TABLE_P2-1_freeze_manifest.md`. AutoResearchBench is pinned to upstream commit `a46c9bfb8968786f73f0a6a5b365b5384cd0f96d`; the released 1,000-record bundle is split by its explicit task-type field into 600 Deep and 400 Wide records before candidate custody. The Wide scorer is deterministic and credential-free. Deep official scoring uses the released title-identity judge and still requires an OpenAI-compatible endpoint; the repository workflow now records this authority precondition without exposing credential values. The credential-free search-time self-exposure diagnostic over the frozen Wide candidate run is `evidence/external_results/AUTORESEARCHBENCH_SEARCH_CONTAMINATION_V1.json`.

The completed MetaSyn probe is archived as `evidence/external_results/METASYN_ID_ONLY_PROBE_V1.json`. It binds all 86 released test reviews, the pinned upstream revision, dataset revision, candidate-output tree and official ID-only evaluator output. The evaluator-side integer error ledger is separately archived as `METASYN_SCREENING_FN_LEDGER_V1.json`; its regeneration reloads the pinned gold only after candidate output is frozen and verifies the official per-review metrics before emitting exact retrieval and screening false-negative counts. These artifacts make the bounded MetaSyn retrieval/screening result reproducible without reclassifying it as a result for the full system.

The public screening transport successor is archived under `development/p2-zenodo-transport-successor-2026-08-23/`. The exact source identity and licence are in `SOURCE_FREEZE.json`; the accidental first-row label exposure is preserved in `SCHEMA_PREFLIGHT_INCIDENT.md`; and the failed V1 binding result remains `RESULT_V1_RETAINED.json`. The post-schema repair is frozen separately in `PROTOCOL_FREEZE_V2.json` and `IMPLEMENTATION_FREEZE_V2.json`; the componentwise estimand and non-transported coordinates are explicit in `P2_ONE_POOL_TRANSPORT_ESTIMAND_AMENDMENT_V1.json`. The executable ordering is `run_transport_successor_v2.py`, and the result and human-readable boundary are `RESULT_V2.json` and `RESULT_REPORT_V2.md`. The public source body is not committed; it is retrievable as Zenodo record 10423427 under CC

BY 4.0 and is accepted only at the frozen byte count and MD5.

The post-V2 active-comparator audit is archived separately under `development/p2-zenodo-active-comparator-audit-2026-08-23/`. Its protocol and implementation were frozen before either comparator outcome; the pinned ASReview archive and selected donor source files are hash-bound in `PROTOCOL_FREEZE_V1.json`. The cadence-matched runner and immutable negative receipt are `run_active_comparator_audit_v1.py` and `RESULT_V1.json`. The audit does not redistribute the Zenodo body or the pinned donor archive and does not claim an exact ASReview application replay.

The cross-review SWIFT V1–V4 records are archived under `development/p2-swift-cross-review-controller-transport-2026-08-23/`. The retained population-mechanics failures are `RESULT_V1.json`, `RESULT_V2.json`, `POPULATION_BINDING_FAILURE_DIAGNOSIS_V1.json` and `POPULATION_BINDING_FAILURE_DIAGNOSIS_V2.json`. V3 is bound by `PROTOCOL_FREEZE_V3.json`, `IMPLEMENTATION_FREEZE_V3.json`, `BINDING_RECEIPT_V3.json` and `RESULT_V3.json`; its human report and failure provenance are `RESULT_REPORT_V3.md` and `FAILURE_ATLAS_V3.json`. The post-outcome diagnostic-only freeze is `NEXT_CONTROLLER_SUCCESSOR_PROTOCOL_V4.json`. The local PubMed title/abstract snapshot is not redistributed: it is 90,520,080 bytes with SHA-256 `962aeb56416dc72fe54336eb4867d14e17505b9b28cb7baf2fd469e82439a568`. `PUBMED_SNAPSHOT_RECEIPT_V1.json` retains its acquisition receipt, while `SOURCE_PAYLOAD_MANIFEST.json` records the exact hashes and byte counts of all omitted source payloads. The V1–V3 frozen runners and original source payloads remain hash-bound but are not redistributed with the manuscript integration packet.

The content-disjoint SYNERGY V5 successor is archived under `development/p2-swift-content-disjoint-v5-2026-08-23/`. Its before-outcome ordering receipt, population/content freeze, protocol and implementation freezes, result, failure atlas and independent arithmetic/hash audit are `PREREGISTRATION_BEFORE_OUTCOME_RECEIPT_V5.json`, `POPULATION_AND_CONTENT_FREEZE_V5.json`, `PROTOCOL_FREEZE_V5.json`, `IMPLEMENTATION_FREEZE_V5.json`, `RESULT_V5.json`, `FAILURE_ATLAS_V5.json`, and `SCIENTIFIC_VERIFICATION_V5.json`. The 12 publisher files (287,754,769 bytes) matched their publisher SHA-1 and local SHA-256 values but are not redistributed. No title, abstract, label or active ordering is stored in the integration packet.

The outcome-unopened KIFMS V6 source/protocol packet is archived under `development/p2-continuous-recall-effort-v6-2026-08-23/`. The source rights and exact 14-file revision-one identity are `SOURCE_RIGHTS_PROVENANCE_RECEIPT_V6.json`; the label-blind 5,074-to-4,934 population calculation and its one raw V5 match are `LABEL_BLIND_OVERLAP_RECEIPT_V6.json`; and the population and continuous estimand freezes are `SOURCE_FAMILY_AND_POPULATION_FREEZE_V6.json` and `PROTOCOL_FREEZE_V6.json`. The feasibility result, negative ledger and integrity receipt are `SOURCE_FEASIBILITY_RESULT_V6.json`, `NEGATIVE_RESULT_LEDGER_V6.json` and `SCIENTIFIC_VERIFICATION_V6.json`. The final packet manifest has SHA-256 `cb7aac3ea6cc4b1070515dff882c4e3cb0a85632e5829af379208dd70448f710`; the protocol has SHA-256 `55a5363e6f9360c4722979d0cb11751575f050ce2eea2147a7883d679ea09633`. The 14 CC-BY-4.0 source CSVs and comparison source bodies are retrievable at their receipt-bound interfaces but are not redistributed in this packet. All KIFMS PubMed-identifier values are empty, so content hashes, not PMID matches, establish the final disjoint population. Historical public ASReview simulation artifact paths exist for these datasets; their metric contents and all exact V6 labels, classes, seeds, rankings and comparative scores remained unopened. Consequently the packet is not evidence of a globally unstudied family, independent custody or comparative performance.

The subsequent transparent public-development execution is archived under `development/p2-kifms-transparent-execution-v7-2026-08-23/`. Its protocol and implementation hashes were frozen before any KIFMS label, class count, seed, order or V7 metric was opened. `RESULT_V7.json` retains all four complete rankings per review through hashes and metrics but no title, abstract or row-level label; `RESULT_REPORT_V7.md` and `NEGATIVE_RESULT_LEDGER_V7.json` give the scientific diagnosis, and the packet-native validator checks the fixed hashes, 4,934-row population, both classes in 14/14 reviews, gate arithmetic and 14-file manifest. Exact public source CSVs remain outside the repository. V7 explicitly records `independent_custody=false`; it is not confirmation.

The nested donor-envelopment development successor is archived under `development/p2-donor-envelopment-v8-2026-08-23/`. The protocol, implementation freeze, exact result, adverse scientific adjudication, recursive ledger and next discriminator are retained separately because the raw execution selector activated two residuals whereas the cross-fitted scientific result admits none. `SCIENTIFIC\_ADJUDICATION\_V8.json` is authoritative for the adverse terminal; the packet validator checks 14 exact u4 reproductions and 84 residual executions. No public source text or row-level label is redistributed.

The source-disjoint title-emphasis stop and its provider-native identity successor are archived under `development/p2-source-disjoint-title-emphasis-v9-2026-08-23/` and `development/p2-title-emphasis-conflict-resolved-v10-2026-08-23/`. V9 retains the single conflicting-label population stop and executes no model arm. V10 retains both provider-native nonduplicate records, executes the frozen three arms once on 21,897 rows, and records the failed coprimary magnitude and sign gates. Its result SHA-256 is `c69e5634b8d0e82a1fa393dbae276d728e9a2f7ba4c9db81a27c0329e2e66742`. No source text or row-level labels are redistributed, and neither packet is independent confirmation.

The conditional safe-residual and donor-saturation theory is archived under `development/p2-safe-residual-envelope-theory-v11-2026-08-23/`. The theory, finite witnesses, JSON claim ledger and receipt are bound by its `SHA256SUMS`; `THEORY.md` SHA-256 is `a3c9107ad152766efc2a08d722657d550781f1f181f47f3ceeba041c0a710176`. V11 performs no empirical or model run and does not establish that an unseen positive family exists.

The source-state successors are archived under `development/p2-state-expanding-acquisition-v12-2026-08-24/` through `development/p2-state-expanding-acquisition-v15-2026-08-24/`. Their `SHA256SUMS` files have SHA-256 values `f2a5cb0fb214d84807ee0396808edb36b619b59ca2066900a6026e163b98cef0`, `e01b9e6bf2e2169d318455a95e1b642a403e228b159eadb7d70295cb73e2c86e`, `8bb23e1c91d59328b64c70fc04af39840c78830f4d4cea1d588f2698495234e6`, and `c8bcfeafea18b06d58ab740da508a6d323de1a1b1ce946e43e0f79df1b114047`, respectively. The corresponding result SHA-256 values are `99c054ac292a7ceb2602323ca011f651f8f3ca3f00fb1a0d25b2684979836ff4`, `8f8681febfccc390f8e3392d321e01737c0ef4cfd2f4f3db45ab8410a15f2cbf`, `95496f053d762173285b9fbb436ff19bad08f47a82e278bb44715b565a0cd333`, and `2c394297445ad68019e7f559697d4141f5930532aa6dfef2e1525687bfb07107`.

V12 retains the failed exact-V10-identity gate and its post-gate diagnosis; V13 retains the positive two-record non-simulation witness and four-of-seven support ceiling. V14 records the frozen-commit index mismatch and the decision not to substitute later commit `dc2dadfdbb98eb1b4259604789abd640aa3b693e`. V15 binds the signed commit/tree/tag, 61-blob dataset manifest and exact historical MIT licence while keeping current CC0 metadata separate. Its unexecuted template error and the successor-only restored V14 route are recorded in `IMPLEMENTATION_CORRECTION_V15B.json`, SHA-256 `55dda1c29a0e0a619f7234888ffdf0d20a2e35d4b8782972343254d694607980`. The outstanding three-role custody handoff is `INDEPENDENT_SOURCE_CUSTODY_REQUEST_V15.json`, SHA-256 `f07c8185c48d5e0a40d02e7bee4dcc0530922fe68e111c0a4cbd83c7c36e2a58`. No packet contains an index parse, review census, model ranking or performance metric that can promote the manuscript terminal.

The OpenAlex V1 development trace received a rights-preserving archival redaction after an adversarial audit found decrypted `Q0_RAW` strings in two duplicate in-tree traces. Original bytes with SHA-256 10ffd34a8b3de5e3b9e45b0e309bf451ff201c2dd0c5d1837e70504959d684cd are quarantined outside the redistributed worktree at mode 0600. The in-tree copies omit the plaintext and retain only per-task SHA-256 provenance; their redacted SHA-256 is 3eaee226dfb323ad812b67b533be70041345b734e54a447fddedf9f1c2881797. The machine-readable mapping is development/p2-openalex-public-successor-2026-08-23/Q0_RAW_QUARANTINE_RECEIPT_V1.json. Authoritative upstream permission for plaintext redistribution remains CANNOT-CHECK; no plaintext circulation is authorized and this archival action is not a legal opinion.

All three OpenAIRE/Crossref Wide campaigns are preserved rather than left behind an expiring Actions URL. Their exact results and run receipts are stored under evidence/external_results/, one prospective freeze per campaign under protocol/; checksummed mirrors of the shared public candidate capture, the failed first evaluator handoff, and the successful evaluator-only repair are under evidence/ci_mirror/. The mirror manifest records both archive-level and per-file SHA-256 digests. Evaluator gold never entered any candidate capture. The first campaign is undetermined because 400 of 1,200 planned provider calls failed; the second because its transport probe returned an empty result array below its frozen minimum, so acquisition never began; the third because provider validity reached 0.716 against a frozen 0.90. None is undetermined because evidence was discarded.

The prospective freeze for the successor acquisition mechanic is protocol/P2_LEXICAL_ECHO_SUCCESSOR_FREEZE_2026-08-21.json, written before any corpus was generated; the mechanic itself is registered there under the identifier D4_DISCRIMINATIVE_TERM_GATING.

The resolution audit that establishes the measurement-floor finding is under evidence/audit/. It reads the published campaign results and reports whether each could have produced a different verdict; it is reproducible from the repository and carries, in the same artifact, a control campaign with resolution so that a check firing on every input can be told from a finding.

The live-provider capture implementation and campaign manifest retain request URL/parameters, query derivation, timestamp, raw body, hash and typed transport status. The frozen live protocol requires arXiv plus two OpenAlex-backed route families. A non-leaking Actions preflight is archived as evidence/access/LIVE_PROVIDER_PREFLIGHT_2026-08-16.json; it confirms that arXiv is credential-free but that no funded OpenAlex API key is available in the same-repository execution context. Therefore the three-route live campaign cannot be frozen for execution here, and no raw live campaign or live monetary-cost ledger is fabricated. Operator re-scope 2026-08-17 removed the empirical-cost requirement: the sanctioned cost axis is the host-recorded search-query count (Figure 3, protocol/TABLE_P2-2_baseline_ablation_cost.md), not a live OpenAlex or arXiv API bill.

SIEVE external-arm preflight

The exact candidate identity and terminal-preserving adapter requirements are recorded in protocol/P2_SIEVE_ADAPTER_PREFLIGHT_V1.json. They bind SIEVE/SkimSearchAgent to the paper-linked Apache-2.0 repository revision 4a640296770c39b8274e7678c6de844daea9541c and prohibit its native answer, empty-answer, error or timeout states from acquiring P2 closure authority. The bounded one-instance stub fixture and its trace/result hashes are archived under development/comparator-binding-harness-2026-08-23/; they establish only that the pinned entrypoint can run. The current status is P2_SIEVE_RUNNABLE_FIXTURE__NOT_A_FROZEN_ARM: no archived P2 task-world reproduction, matched provider/route/budget binding, or comparator performance result is supplied by this preflight.

15 Reproducibility

Four levels are distinguished, and we claim only what the current artifacts support.

Fully reproducible without network access. The 390-task offline controlled-index companion regenerates from committed data and subject code. A clean GitHub Actions job independently rebuilds the 16,380-run publication projection and compares it to the committed summary. The frozen plan awards the result its committed precision tier together with a mandatory underpowered label on the primary offline comparison; reproduction does not elevate that authority.

Reproducible bounded external evidence. The pinned MetaSyn workflow reconstructs the public/gold split, runs the credential-free BM25 plus public-protocol screening candidate, executes the official ID-only evaluator, and can regenerate the exact false-negative ledger. This is external evidence for that declared probe only.

Runnable credential-free benchmark lanes. AutoResearchBench Wide scoring is deterministic once the gold-blind candidate output is frozen. The same public-arXiv candidate machinery also supports the declared deterministic Deep exact-ID deviation. The official Deep title judge remains a distinct authority and is not replaced by the ID deviation when no judge endpoint is available.

Executed official Deep judge archive. The bounded keyless public-arXiv probe of the official Deep LLM title judge is archived as `evidence/external_results/DEEP_OFFICIAL_ARCHIVE_V1.json` with its nine-case positive/negative judge control in `DEEP_JUDGE_CONTROL_2026-08-17.json`. The checkpointed judge runner and the archive finalizer are both reproducible from the repository as `scripts/judge_checkpointed.py` followed by `scripts/finalize_deep_scoring.py`; the archive records the judge model, the pinned prompt digest, the run manifest hashes, and the bounded probe scope label.

Capture-capable but blocked as a final live campaign. The committed live layer is designed to archive mutable provider evidence rather than re-derive it, but the frozen route set requires funded OpenAlex access that is absent in the verified execution context. Until that authority is supplied and the campaign is actually run, raw live-provider outcomes, empirical live monetary cost and wall-clock, and the matched external discovery claim all stay undetermined. Query-count metrics from the offline companion are the completed cost evidence in this revision.

16 Ethics

This work does not collect new human-subject records or personally identifiable information. The frozen complete-gold companion is fully synthetic and regenerable from a recorded seed. External probes use publicly released benchmark questions and evaluator interfaces; third-party corpora are not redistributed where the access-audit table records a licence block or partial redistribution. Dual-use risk is that of any scientific-literature retrieval system: the paper evaluates discovery governance and does not present a deployment, surveillance, or human-facing study.

17 Conclusion

Scientific discovery systems should be compared as joint acquisition–authority controllers, not as a ranker followed by an unexamined stop decision. The envelope formalism makes that comparison donor-monotone and non-compensatory: stronger retrieval, reasoning, coverage, stopping, obligation and memory methods are absorbed into the comparator class, while false closure cannot be traded for report quality or lower cost. Its registered product-order rule requires guard passage, coordinatewise no-harm and a prefrozen strict margin against the complete donor frontier before claiming superiority. The acquisition-ceiling theorem attributes failures before proposing a repair; the factorization theorem states exactly when a product has enough state to implement scientific closure; the indistinguishable-world bound proves when new acquisition or new assumptions are unavoidable; and the robust safe-set rule is maximally permissive, while the registered obligation rule reaches it only under an explicit completeness condition.

The evidence is deliberately heterogeneous rather than averaged into a single claim. The controlled-index campaign supports mechanism behavior but remains underpowered for its frozen superiority margin. The external 7/229 exposure result is an adverse candidate-generation diagnosis, and the provider-invalid matched campaign remains undetermined. A source-disjoint public screening pool does show a large conditional within-pool contrast after reuse of the same active update rule, hyperparameters and batch-size rule under a source-specific representation and label-informed warm start—0.703409 recall at 10% screened versus 0.335989 for the better-performing of two registered passive within-pool baselines. A post-V2 active audit found 0.722444 for cadence-matched pinned ELAS u4, so the candidate did not beat the stronger active comparator. A subsequent five-review SWIFT successor made the adverse boundary more general within public development: on 96,241 canonical rows, candidate-minus-u4 unweighted review-level mean recall@10% and WSS@95 were -0.021717971642 and -0.050396333672, respectively, and u4 was better on both endpoints in every review. The primary and relative-work-saving gates failed; harm and absolute work saving passed without compensation. Thus that one-pool post-schema-repair result is not a population transport estimate and belongs below the open-world acquisition and authority claims. The exact-contract battery supports the predicted authority distinction—400/400 against 250/400 for a product that drops unresolved material routes—while the information-equivalent product's 400/400 tie confirms that the residual is state sufficiency, not inherent centralized expressivity. These are components of a wider theory, not evidence of deployed general superiority.

The upward research target is now precise rather than merely narrower: exceed the guarded donor-complete envelope on source-disjoint naturalistic scientific worlds by inventing routes that lift the acquisition ceiling, preserving decision-relevant future options, and closing safely under independently held obligation custody. The registered 768-cluster, four-arena target is designed to test that claim, but its worlds and comparator identities are not frozen for execution; it has not run and cannot change the current terminal. The post-outcome SWIFT V4 representation-by-learner/balancer factorization selects the learner/balancer component as a stable within-panel diagnosis of the adverse V3 gap. It promotes no arm on SWIFT; the selected mechanism requires a content-disjoint, outcome-unopened review family with unchanged gates. The prospectively frozen SYNERGY V5 successor supplies that content-disjoint test over five review decisions and 101,044 canonical records, but the selected component fails its locked recall@10% magnitude and sign gates: its mean effect is $+0.008834277869$, strictly positive in three of five decisions and exactly zero in two. Positive recall@5%, recall@20%, and WSS@95 diagnostics do not replace the failed primary rule, while the full candidate-minus-u4 arm remains adverse at recall@10% and WSS@95. The next test must therefore use a new source family, retain recall@10% as an adverse coprimary endpoint, and freeze a continuous recall–effort estimand before outcomes. V6 now binds that source/protocol layer without opening a KIFMS label or executing a comparison: 14 CC-BY-4.0 OSF review decisions yield 4,934 label-blind canonical rows after one raw V5 match and all within-family shared content are removed; final SWIFT and V5 content overlap is zero. The normalized recall–effort area over the first 20% screened is coprimary with the unchanged adverse recall@10% rule, and u4 plus every harm/work-saving gate is retained. All declared PMID values are empty, so only title–abstract content identities establish disjointness. Historical public ASReview artifact paths also mean that this is exact-V6 outcome non-access, not a globally unstudied family. Independent protected custody remains undetermined. V7 nevertheless removes the execution wait under a new, explicitly nonconfirmatory identity: all 14 exact files and the 4,934-row population reproduce, every review has both classes and all four rankings complete. Mean learner/balancer effects are +0.084472 on CRE20, +0.148820 on recall@10% and +0.119813 on WSS@95, but CRE20 and recall@10% are strictly positive in only 11/14 and 9/14 reviews, and the worst-review recall@10% effect is −0.125. The full candidate-minus-u4 means are −0.201648 at recall@10% and −0.119160 on WSS@95. Thus six locked gates fail and no arm is promoted. The upward successor must envelop the strong u4 donor and learn only a cross-fitted residual with a hard u4 fallback, then test once on a new source-disjoint family; the V5/V7 adverse results and current manuscript terminal remain unchanged. V8 performs that residual search only on now-open KIFMS development data. It reproduces exact u4 in 14/14 reviews, tests 84 residual executions, falls back to u4 in 12/14 outer folds and observes cross-fitted deltas of −0.002009 CRE20, −0.028274 R@10 and −0.000991 WSS95, with worst-review R@10 −0.333333. Both activated residuals harm their held-out review and zero of six full configurations jointly passes CRE20, WSS95 and harm support. The adjudicated terminal therefore admits no residual and preserves exact u4 fallback; title emphasis is retained only as one bounded diagnostic for a not-yet-bound source-disjoint family. V9 reaches that family but stops before fitting because one normalized content identity has conflicting labels. V10 freezes the provider-native nonduplicate disposition, retains both records, and executes once on 21,897 rows. Mean deltas versus exact u4 are +0.000780 CRE20, −0.005383 R@10 and +0.001765 WSS95; positive CRE20 and R@10 signs occur in 4/7 and 1/7 reviews. Four frozen gates fail, so title emphasis is discarded without tuning and exact u4 remains the fallback. V11 then derives the wider positive boundary: a certificate-switched residual envelope embeds the donor, finite failures do not identify universal donor optimality, and strict same-contract separation from a saturated donor class requires fibre-separating information or state, new acquisition support, or proof that the class was incomplete.

V12–V15 instantiate that boundary at the source layer. Provider-native keywords can separate records that exact u4 model text cannot, and the coherent signed snapshot exposes 61 dataset blobs under an exact historical identity. These findings establish available state and a bound source frontier, not a screening benefit; population binding, matched outcomes and independent custody remain necessary before the narrowed terminal recorded for this work can be raised.

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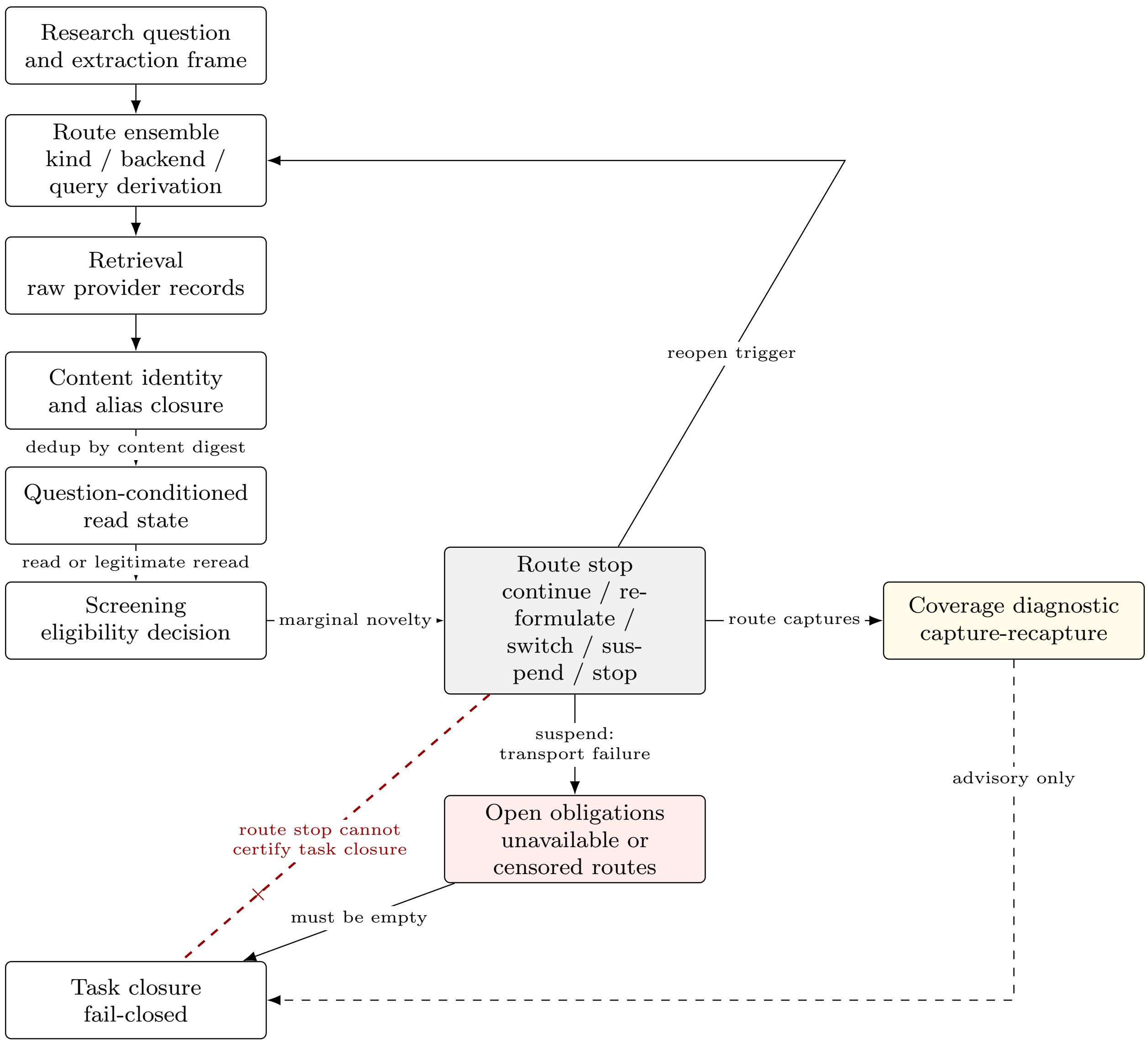

Figure 2: The discovery pipeline, separating retrieval, screening, route stopping and task closure. Solid arrows are authority-bearing transitions; the dashed arrow marks the coverage diagnostic, which informs but never certifies closure; the crossed arrow marks the transition the design forbids, namely a route stop certifying task-level completeness (Proposition 2). A route suspended by transport failure leaves an open obligation that closure must clear, so provider unavailability cannot become evidence of absence. The figure is generated from the same specification the tests read, so it cannot drift from the state machine it illustrates.

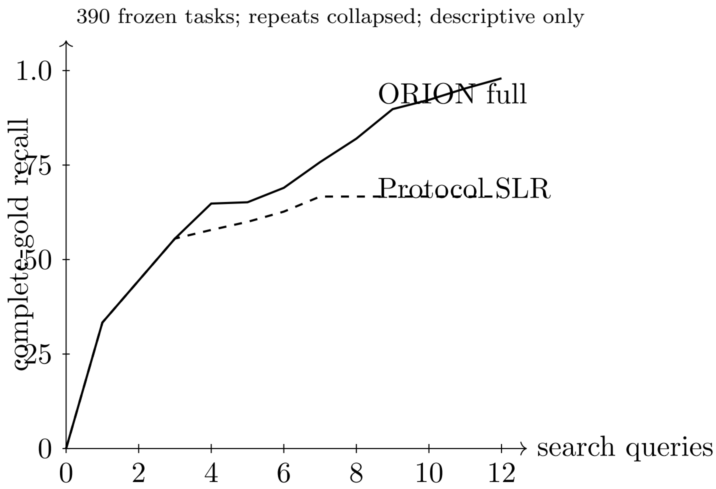

Figure 3: Cumulative complete-gold recall versus the host-recorded number of search queries for the governed system and the strongest frozen confirmatory baseline. Deterministic repeats are first checked for identical traces and then collapsed within task; the curves are means over the 390 tasks and are descriptive only. Synthetic wall-clock is intentionally fixed to zero, so the controlled companion supports the query axis, not an empirical-time claim.

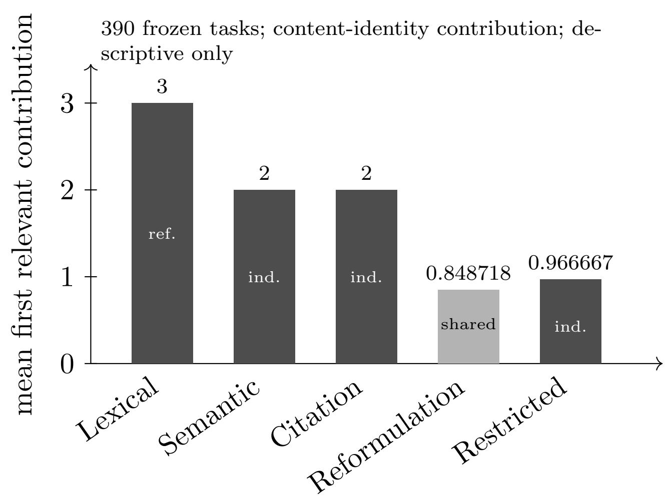

Figure 4: Mean number of first relevant content identities contributed by each route in the governed system's campaign order. "Independent" is an earned structural annotation relative to the lexical reference route; the reformulation route is deliberately marked shared-backend even though it adds a new identity. Values are task means over the frozen descriptive companion.

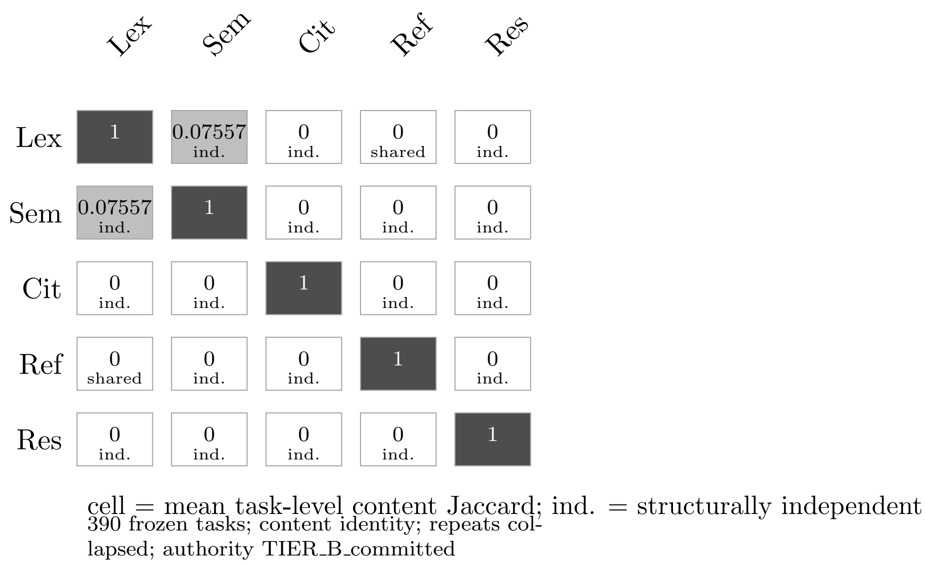

Figure 5: Mean task-level Jaccard overlap between route captures, computed over content identities rather than source locators, together with the structural independence verdict for each off-diagonal pair. The lexical–semantic overlap is 0.07557; the other off-diagonal mean overlaps are zero in this controlled world. Zero observed overlap is not itself evidence of independence.

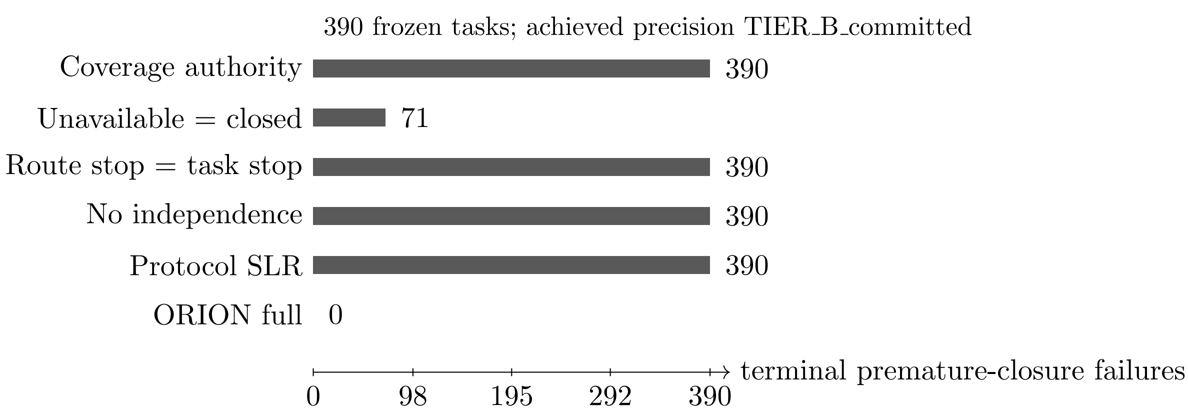

Figure 6: Terminal premature-closure classifications for selected stopping-safety controls in the frozen 390-task offline companion. The plot is generated from the immutable publication summary. It shows task counts, not an inferential estimate; lower-precedence failure mechanisms can be masked by the evaluator's terminal-failure precedence.